Universidad Complutense de Madrid

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Introduction to Morse Theory

Bachelor's Thesis

Degree in Mathematics

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A la memoria de Bimba. Gracias por acompañarme hasta casi el final de este camino.

Resumen

La teoría de Morse permite extraer de una variedad información sobre sus invariantes topológicos mediante el estudio de los puntos críticos de funciones diferenciables con valores reales.

Dada una función diferenciable $f : M \to \mathbb{R}$ sobre una variedad M, el lema de Morse establece que alrededor de todo punto crítico no degenerado existe un sistema local de coordenadas respecto del cual f se expresa como una forma cuadrática. Esto tiene como consecuencia que los puntos críticos no degenerados son aislados y constituye el primer pilar de la teoría de Morse, encargada de estudiar precisamente las funciones cuyos puntos críticos son todos no degenerados. Estas funciones se conoces como funciones de Morse.

La importancia de la teoría de Morse radica en que permite determinar la topología de una variedad a partir de los puntos críticos de una función de Morse. Concretamente, si los conjuntos subnivel de una función f de Morse son compactos, se puede relacionar el tipo de homotopía de la variedad con el de un CW-complejo con tantas celdas de dimensión λ como puntos críticos de índice λ tenga f. Por otro lado, las desigualdades de Morse determinan cotas para los números de Betti de una variedad y permiten computar su característica de Euler-Poincaré en función únicamente del número de puntos críticos de cada índice posible de una función de Morse.

Abstract

Morse theory allows us to extract from a manifold information about its topological invariants by studying the critical points of real-valued differentiable functions.

Given a differentiable function $f : M \to \mathbb{R}$ on a manifold M, Morse's lemma establishes that around every non-degenerate critical point there exists a local coordinate system with respect to which f is expressed as a quadratic form. This, in turn, implies that non-degenerate critical points are isolated and constitutes the foundation of Morse Theory, which studies functions whose critical points are all non-degenerate. These functions are called Morse functions.

The importance of Morse theory lies in the fact that it allows one to determine the topology of a manifold from the critical points of a Morse function. In particular, if all sublevel sets of a Morse function f are compact, it is possible to relate the homotopy type of a manifold with that of a CW-complex with as many cells of dimension λ as critical points of index λ f has. In addition, Morse inequalities determine bounds for the Betti numbers of a manifold and let us compute its Euler-Poincaré characteristic relying only on the number of critical points of each possible index of a Morse function.

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Introduction

The journey of Morse theory begins in the 1920s, when a young Marston Morse, immersed in Poincaré's "Analysis Situs", publishes a paper that relates for the first time the critical points of a differentiable function with the topology of the space on which it is defined [Bot80]. From then on, numerous mathematicians would put their minds at the service of the development of this theory, giving rise to what we know today as Morse theory. We will not provide an exhaustive list, but there are some whom we will indeed mention: in the first place, John Milnor, who in 1959 uses Morse theory to prove the existence of the first *exotic sphere* [Mil10]; and in the second place, Stephen Smale, who proves the *Poincaré conjecture* true for dimensions greater than four and extends the techniques used to prove the *h-cobordism theorem* [Bot88].

In 1963, Milnor takes on the task of compiling in a more modern language the foundations of Morse theory; this work crystallizes in his book "Morse Theory" [Mil63]. In this book he presents a multitude of results within a reduced space, however, this comes at the price of leaving out technical aspects necessary for an inexperienced reader to understand them. In the present work, we seek to present the foundations of Morse theory, from the Morse Lemma (1.2.8) to the Morse inequalities (4.2), in a modern and comprehensive language, and including as much theory as necessary to make it as self-contained as possible.

As we have already mentioned, Morse theory relates the critical points of a differentiable function with topology. The critical points of a function are classified according to whether their Hessian matrix is singular or not at them, being called in the second case *non-degenerate critical points*. Non-degenerate critical points have the particularity that in a neighborhood of theirs there exists a local coordinate system with respect to which the function is expressed as a quadratic form; which implies that they are isolated.

A real-valued function on a smooth manifold whose critical points are all non-degenerate is known as a *Morse function*. By studying the sublevel spaces $M^a = \{q \in M | f(q) \leq a\}$ of a Morse function $f : M \to \mathbb{R}$ one can learn how the topology of M changes when passing through the critical points. Theorem 2.2.17 establishes that if all the sublevel sets are compact, then the homotopy type of the manifold is that of a CW-complex with as many cells of dimension λ as critical points of index λ f has. Moreover, the Morse inequalities make use only of the number of critical points of each possible index to determine the Euler–Poincaré characteristic and bound the Betti numbers of the manifold.

In essence, Morse theory allows us to know a great amount of information about the topological invariants of a manifold knowing only indices of critical points. Procedures that previously required complex arguments and algebraic techniques are reduced to basic techniques of analysis.

Returning to the results mentioned above, Milnor constructs a manifold equipped with a differentiable structure and proves that it is homeomorphic but not diffeomorphic to the 7-dimensional sphere $\mathbb{S}^7$. A manifold of this kind is known by the name of exotic sphere, and the way in which he proves that it is homeomorphic to $\mathbb{S}^7$ is by defining on it a Morse function with only two critical points [Mil10] (see 2.2.20). The case of the h-cobordism theorem is more complicated, but let us say that its proof begins with the definition of a Morse function on the cobordism and, making use of it, gives its decomposition into handles [Mil25].

An example with which to illustrate Morse theory is the following. Suppose we have the torus resting on a plane as in Figure 1, and we define on it the function $f : T^2 \to \mathbb{R}$ that takes each point to its height with respect to the plane.

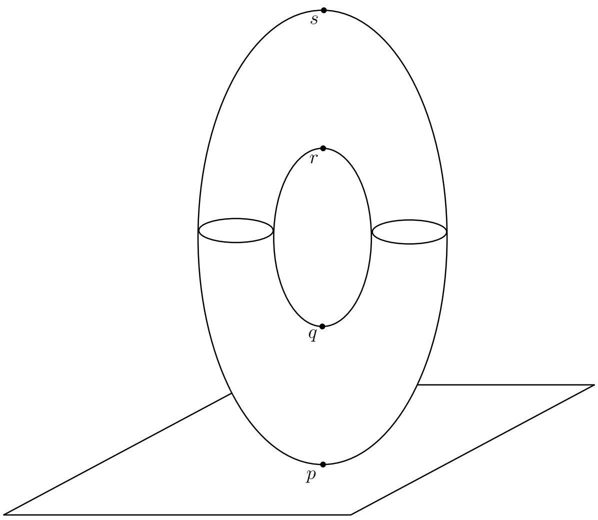

Figure 1: Torus resting on a plane.

Well, f is a Morse function with four critical points, a minimum (p), a maximum (s) and two saddle points (q and r). Let $a \in \mathbb{R}$, the following holds:

(i) If $a < f(p)$, M^a is empty.
(ii) If $f(p) < a < f(q)$, then M^a is homeomorphic to a 2-disk.
(iii) If $f(q) < a < f(r)$, we have that M^a is homeomorphic to a cylinder.
(iv) If $f(r) < a < f(s)$, M^a is homeomorphic to a compact manifold with a circle as boundary.
(v) If $f(s) < a$, M^a is the full torus.

Let us call these regions, in order, M^{a_0}, M^{a_1}, M^{a_2}, M^{a_3} and M^{a_4}. In order to describe how the topology of M^a changes as it passes through each critical point, it is necessary to know their indices. The indices of p and s are 0 and 2 respectively; the indices of q and r are both 1.

Now, in terms of homotopy, M^{a_1} is indistinguishable from a point. When passing to M^{a_2}, the homotopy type changes to that of a 2-disk with a 1-cell attached, the same as that of the cylinder. When we pass to M^{a_3}, once again the change in homotopy type corresponds to attaching a 1-cell. Finally, when we reach M^{a_4}, a 2-cell is attached and one has the homotopy type of a CW-complex with two 2-cells and two 1-cells. That is, with this sequence of steps we have recovered the homotopy type of the torus [Mil63] [Bot88].

This result is not particular to the torus in this position, it is general: Morse theory allows us to reproduce this same process for any manifold. We will prove that the topology of the sublevel sets changes only when passing through a critical point. In particular, we will prove that if M^a and M^b are such that there exists no critical point $p \in M$ such that $a < f(p) < b$ and $f^{-1}[a,b]$ is compact, then M^a is a deformation retract of M^b and, in fact, they are diffeomorphic (Theorem 2.2.5).

In the same way, we will prove that if there exists a critical point $p \in M$ satisfying $a < f(p) < b$, then the change in topology is, concretely, the one produced by attaching a λ-cell to M^a, where λ is the index of p (Theorem 2.2.7).

In the first chapter we will introduce the notion of non-degenerate critical point and prove the Morse Lemma. In the second chapter we will give a brief formal introduction to CW-complexes and prove the results that describe the procedure used in the example above. The third chapter will be exclusively devoted to developing homology theory, giving constructions of simplicial, cellular and relative homology and presenting the Betti numbers and the Euler–Poincaré characteristic. The fourth and last chapter will be about the Morse inequalities, with a first exposition of certain results on subadditive functions.

Chapter 1

Morse functions

1.1 Preliminaries and notation

To eliminate possible ambiguities in the use of certain terms, we begin by presenting smooth manifolds and establishing the direction of the charts that will be used from now on. It is also worth paying attention to what we mean when we speak of a local coordinate system and how we denote it.

We understand by *n-dimensional smooth manifold* a pair $(M, \mathcal{A})$ where M is an n-dimensional topological manifold (that is, a Hausdorff, second countable and locally Euclidean topological space) and $\mathcal{A}$ an atlas of M.

In turn, an atlas of an n-dimensional topological manifold M will be a family $\mathcal{A}$ of pairs

$$\{(\mathcal{U}_\alpha,\ \varphi_\alpha : \mathcal{U}_\alpha \to \mathcal{V}_\alpha \subset \mathbb{R}^n)\}_{\alpha\in\Lambda}$$

which we will call *charts*, where the $\mathcal{U}_\alpha$ form an open cover of M and the functions φ_α are homeomorphisms such that the compositions

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(\mathcal{U}_\alpha \cap \mathcal{U}_\beta) \to \varphi_\beta(\mathcal{U}_\alpha \cap \mathcal{U}_\beta)$$

are of class C^∞ for any α and β.

Every space M will denote an n-dimensional smooth manifold unless otherwise specified.

We will frequently speak of a *local coordinate system in a neighborhood $\mathcal{U}$ of $p \in M$*. By this we will mean nothing other than the choice of a chart. We will denote it $(x_1, \dots, x_n)$ and it will be implicit that

$$x_i = \pi_i \circ \varphi_\alpha : \mathcal{U} \cap \mathcal{U}_\alpha \to \mathbb{R},$$

where α will depend on the point of $\mathcal{U}$ on which x_i acts and the chosen chart containing it. In simpler terms, x_i takes a point and sends it to the projection onto the i-th coordinate of its image under one of the charts.

We will abuse notation to refer interchangeably by f to a function $f : M \to \mathbb{R}$ and its expression in coordinates $\tilde{f} = f \circ \varphi_\alpha^{-1} : \mathbb{R}^n \to \mathbb{R}$. That is, we will speak of the image of a point p under f as $f(p)$ and as $f(x_1, \dots, x_n)$, where $(x_1, \dots, x_n) = (x_1(p), \dots, x_n(p))$ are its coordinates in a chosen chart.

On the other hand, a *differentiable map* $f\colon M \to N$ between two smooth manifolds is a continuous map such that, for every pair of charts $(\mathcal{U}, \varphi\colon \mathcal{U} \to \mathcal{V}) \in \mathcal{A}$ and $(\mathcal{U}', \varphi'\colon \mathcal{U}' \to \mathcal{V}') \in \mathcal{A}'$, the map

$$\varphi' \circ f \circ \varphi^{-1} \ : \ \varphi\big(f^{-1}(\mathcal{U}')\big) \subseteq \mathcal{V} \longrightarrow \mathcal{V}'$$

is of class C^∞. In what follows, we will refer to them as *smooth* maps and as *diffeomorphisms* if they possess a differentiable inverse.

The tangent space to M at a point p is defined as

$$T_pM = \{X : C^\infty(p) \to \mathbb{R} \mid X \text{ is a derivation}\},$$

where $C^\infty(p)$ denotes the set of functions differentiable in a neighborhood of p and a *derivation* at p is an $\mathbb{R}$-linear function $X : C^\infty(p) \to \mathbb{R}$ satisfying the Leibniz rule, that is, such that $X(f \cdot g) = X(f)g(p) + X(g)f(p)$ for any $f, g \in C^\infty(p)$.

When we have a differentiable map f we will be able to speak of its *differential at a point* $p \in M$, which will be the linear map

$$\mathrm{d}_p f\colon T_pM \longrightarrow T_{f(p)}N$$

defined by

$$\mathrm{d}_p f(X) = X(\bullet \circ f),$$

where $\bullet$ indicates the position of the argument. Explicitly, the vector $\mathrm{d}_p f(X) \in T_{f(p)}N$ is the derivation that, for every function g differentiable in a neighborhood of $f(p)$, is given by

$$\mathrm{d}_p f(X)(g) = X(g \circ f).$$

Let us observe that this differential will be the usual linear map given by the matrix of partial derivatives of f whenever f is a function from $\mathbb{R}^n$ to $\mathbb{R}^m$.

Given an n-dimensional smooth manifold M we will call the *tangent bundle* of M the disjoint union of the tangent spaces at each of the points of M, that is,

$$TM \ = \bigsqcup_{p \in M} T_pM.$$

The tangent bundle can be endowed with a smooth manifold structure of dimension $2n$ in a manner compatible with the charts of M and such that the map $\pi : TM \to M$ given by

$$\pi(p, X) = p$$

is differentiable.

In turn, a *smooth vector field* on an M is a smooth map

$$X\colon M \to TM,$$

such that $\pi \circ X = \mathrm{id}_M$.

Finally, we will speak of *Riemannian manifolds* when what we have is a smooth manifold M equipped with an *inner product* on each tangent space, that is, a pair $(M, \langle\cdot,\cdot\rangle)$ where M is a smooth manifold and for each point $p \in M$, $\langle\cdot,\cdot\rangle_p$ is an inner product (a bilinear, symmetric, positive definite form)

$$\langle\cdot,\cdot\rangle_p \colon T_pM \times T_pM \to \mathbb{R},$$

satisfying that for any smooth vector fields X and Y, the function

$$p \mapsto \langle X(p), Y(p)\rangle_p$$

is smooth on M. This inner product is known as a *Riemannian metric*. It is worth mentioning that we will sometimes refer to a Riemannian manifold simply as M if we do not need to make explicit use of the metric, since every smooth manifold can be equipped with one.

For greater depth on most of these concepts we suggest turning to [Gon24], from which we ourselves have obtained a great deal of the information.

Now that we have gotten rid of possible gaps as far as smooth manifolds are concerned, let us make our way into *Morse theory*.

1.2 The Morse Lemma

We begin by introducing the critical points of a smooth function and the concept of index of a non-degenerate critical point. We will close the section with the first pillar of Morse theory, upon which a large part of the later results will be built: the Morse Lemma.

Definition 1.2.1. Let f :M→ $\mathbb{R}$ be a smooth function.

i) We will say that $p \in$ M is a *critical point of f* if $d_pf : T_pM \to \mathbb{R}$ is identically zero. That is, given a local coordinate system $(x_1, \ldots, x_n)$ in a neighborhood $\mathcal{U}$ of p we have

$$\frac{\partial f}{\partial x_1}(p) = \cdots = \frac{\partial f}{\partial x_n}(p) = 0.$$

ii) In turn, we will say that $f(p) \in \mathbb{R}$ is a *critical value of f* if $p \in$ M is a critical point.

Definition 1.2.2. Let $p \in M$ be a critical point of f, we will say that p is *non-degenerate* if the matrix $\left(\frac{\partial^2 f}{\partial x_i \partial x_j}(p)\right)$ is non-singular, that is, its determinant is non-zero.

Definition 1.2.3. Let $f : M \to \mathbb{R}$ be a smooth function and $p \in M$ a critical point of f. We define $\mathrm{Hess}_p f : T_pM \to \mathbb{R}$ as the symmetric bilinear form on the tangent space to M at p constructed as follows:

We take $v, w \in T_pM$ and let $\tilde{v}$ and $\tilde{w}$ be local extensions of them to vector fields. Then

$$\mathrm{Hess}_p f(v,w) = \tilde{v}_p(\tilde{w}_p(f)),$$

where $\tilde{v}_p$ is simply v and $\tilde{w}_p(f)$ denotes the partial derivative of f in the direction of $\tilde{w}$.

Remark 1.2.4. It is quick to check that $\mathrm{Hess}_p f$ is indeed symmetric and well defined:

(i) It is symmetric since $\tilde{v}_p(\tilde{w}_p(f)) - \tilde{w}_p(\tilde{v}_p(f)) = [\tilde{v},\tilde{w}]_p(f) = 0$ where $[\tilde{v},\tilde{w}]$ is the Lie bracket of $\tilde{v}$ and $\tilde{w}$, whose expression in coordinates, let us recall, is

$$[\tilde{v},\tilde{w}]_p(f) = \sum_{i=1}^{n} \left(\tilde{v}_p(\tilde{w}_i) - \tilde{w}_p(\tilde{v}_i)\right) \frac{\partial}{\partial x_i}\bigg|_p f.$$

Hence it vanishes since p is a critical point.

(ii) On the other hand, it is well defined since $\tilde{v}_p(\tilde{w}_p(f))$ does not depend on the chosen extensions of v and w, because what matters to us is the vector of the field at the point p and this does not change.

Let us now see what $\mathrm{Hess}_p f$ looks like in coordinates. Again, let us take a local coordinate system $(x_1,\ldots,x_n)$, we have

$$v = \sum_{i=0}^{n} a_i \frac{\partial}{\partial x_i}\bigg|_p \quad \text{and} \quad w = \sum_{i=0}^{n} b_i \frac{\partial}{\partial x_i}\bigg|_p .$$

And for the extensions $\tilde{v}$ and $\tilde{w}$ it suffices to take as coefficients the constant functions a_i and b_j. Thus

$$\mathrm{Hess}_p f = \tilde{v}_p(\tilde{w}_p(f)) = v\left(\sum_{i=0}^{n} b_i \frac{\partial}{\partial x_i}\bigg|_p f\right) = \sum_{i,j} a_i b_i \frac{\partial^2}{\partial x_i \partial x_j}\bigg|_p f.$$

Hence the matrix $\left(\frac{\partial^2 f}{\partial x_i \partial x_j}(p)\right)$ mentioned before represents the bilinear form $\mathrm{Hess}_p f$ with respect to the basis $\left(\frac{\partial}{\partial x_i}\Big|_p, \ldots, \frac{\partial}{\partial x_n}\Big|_p\right)$ of T_pM.

Definition 1.2.5. Let V be a vector space and $H : V \times V \to \mathbb{R}$ a bilinear form on V. Then

i) The *index* of H is the maximal dimension of a subspace of V on which H is negative definite.

ii) The *nullity* of H is the dimension of the radical of H, that is,

$$\dim(\mathrm{Rad}(H)) \quad \text{with} \quad \mathrm{Rad}(H) = \{v \in V : H(v,w) = 0, \ \forall w \in V\}.$$

Remark 1.2.6. A critical point $p \in M$ of a smooth f is non-degenerate if and only if the nullity of $H = \mathrm{Hess}_p f$ on T_pM is 0. Indeed:

$\boxed{\Longrightarrow}$ Suppose there exists $v \in \mathrm{Rad}(H)$ different from zero, then $H(v,w) = v^t H w = 0$ for all $w \in T_pM$, hence $v^t H = 0$ if and only if $v^t = 0$ since H is non-singular and $v^t H$ is a linear combination of the rows of H, contradiction.

$\boxed{\Longleftarrow}$ $\mathrm{Rad}(H) = \{0\}$. Suppose that p is degenerate and let $v \in T_pM \setminus \{0\}$, then there exists $w \in T_pM$ such that $v^t H w \neq 0$. Thus, $v^t H \neq 0$ and since v was arbitrary we have that $v^t H \neq 0$ for all $v \in T_pM \setminus \{0\}$, hence H is non-singular and we have reached a contradiction.

Note. From now on we will refer to the bilinear form induced by f as $\mathrm{Hess}_p f$ and to its matrix with respect to the basis of the tangent space as $H_p f$, or simply H if the context allows it. Moreover, we will refer to the index and the nullity of $\mathrm{Hess}_p f$ as the *index and nullity of f at p.*

Now that we have developed the vocabulary to speak about the critical points of a real-valued function, we close this section with the statement and proof of the *Morse Lemma*.

This result is fundamental for the development of Morse theory. Broadly speaking, it establishes that in a neighborhood of a non-degenerate critical point there exists a local coordinate system with respect to which the function f can be expressed as a quadratic form. From a geometric point of view, this implies that f resembles a paraboloid in said neighborhood. This turns out to be incredibly useful and constitutes the basis of the topological arguments we will use from this point on.

Before stating the Morse Lemma we need to prove the following result from differential calculus.

Lemma 1.2.7. *Let $\mathcal{V}$ be a convex neighborhood of the origin in $\mathbb{R}^n$ and $f \in \mathcal{C}^\infty(\mathcal{V})$ with $f(0) = 0$. Then,*

$$f(x_1, \dots, x_n) = \sum_{i=1}^{n} x_i g_i(x_1, \dots, x_n)$$

for some functions $g_i \in \mathcal{C}^\infty(\mathcal{V})$ which, moreover, satisfy $g_i(0) = \frac{\partial f}{\partial x_i}(0)$.

Proof. We have

$$f(x_i, \dots, x_n) = \int_0^1 \frac{df(tx_1, \dots, tx_n)}{dt} dt = \int_0^1 \left(\sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(tx_1, \dots, tx_n) x_i\right) dt =$$
$$= \sum_{i=1}^{n} x_i \left(\int_0^1 \frac{\partial f}{\partial x_i}(tx_1, \dots, tx_n) dt \right)$$

We take $g_i(x_1, \dots, x_n) = \int_0^1 \frac{\partial f}{\partial x_i}(tx_1, \dots, tx_n) dt$ and it holds that $g_i(0) = \int_0^1 \frac{\partial f}{\partial x_i}(0) dt = \frac{\partial f}{\partial x_i}(0)$.

□

Let us now proceed with the proof, for which we have relied both on Milnor's ([Mil63, p. 6]) and on the one found in [MH93, p. 431].

Lemma 1.2.8 (Morse Lemma)**.** *Let p be a non-degenerate critical point of a smooth function $f : M \to \mathbb{R}$. Then there exists a local coordinate system $(y_1, \dots, y_n)$ in a neighborhood $\mathcal{U}$ of p such that $y_i(p) = 0, \forall i \in \{1 \dots, n\}$ and*

$$f = f(p) - (y_1)^2 - \cdots - (y_\lambda)^2 + (y_{\lambda+1})^2 + \cdots + (y_n)^2$$

holds on all of $\mathcal{U}$ with λ the index of f at p.

Proof of the Morse Lemma. First of all, let us see that if f can be expressed in that manner on $\mathcal{U}$ then λ is necessarily the index of f at p. Let $(x_1, \dots, x_n)$ be a local coordinate system on $\mathcal{U}$ and suppose that

$$f = f(p) - (x_i)^2 - \cdots - (x_\lambda)^2 + (x_{\lambda+1})^2 + \cdots + (x_n)^2.$$

Then we have that

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \begin{cases} -2, \ i = j \leq \lambda \\ \ \ 2, \ i = j \geq \lambda \\ \ \ 0, i \neq j \end{cases}$$

That is, $H_p f$ is a diagonal matrix with λ negative values and $n - \lambda$ positive values. Hence it has λ negative eigenvalues, which means that there exists a subspace of T_pM of dimension λ on which it is negative definite and another of dimension $n - \lambda$ on which it is positive definite.

The only way in which λ could be different from the index of f at p is if there existed a subspace of dimension greater than the one mentioned on which $H_p f$ were negative definite, which is impossible.

Some preparations are needed before proving that the coordinate system $(y_1, \dots, y_n)$ exists.

First of all we may assume without loss of generality that, in coordinates, p is the origin of $\mathbb{R}^n$ by composing the chosen chart with the translation taking p to the origin, which is a diffeomorphism. We also assume $f(p) = f(0) = 0$; if this were not the case we could take $\bar{f}(q) = f(q) - f(p)$, which satisfies $\bar{f}(p) = 0$ and has p as a non-degenerate critical point, and undo the change afterwards.

By the previous lemma $f(x_1, \dots, x_n) = \sum_{j=1}^{n} x_j g_j(x_1, \dots, x_n)$ in a convex neighborhood of the origin. Now, since 0 is a critical point of f we have

$$\frac{\partial f}{\partial x_j}(0) = 0 \ \text{ and } \ g_j(0) = \int_0^1 \frac{\partial f}{\partial x_j}(0) \ dt = 0.$$

Hence we can apply the lemma again to each of the g_j, obtaining

$$g_j(x_1, \dots, x_n) = \sum_{i=1}^{n} x_i h_{ij}(x_1, \dots, x_n).$$

And, therefore,

$$f(x_1, \dots, x_n) = \sum_{i,j} x_i x_j h_{ij}(x_1, \dots, x_n).$$

Moreover, the matrix $H(0) = (h_{ij}(0))$ is equal to $\frac{1}{2}H_0 f = \frac{1}{2}(\frac{\partial^2 f}{\partial x_i \partial x_j}(0))$ and is therefore symmetric and non-singular. Indeed, we know that

$$h_{ij} = \int_0^1 \frac{\partial g_i}{\partial x_j}(tx_1, \dots, tx_n)dt \quad \text{and} \quad g_i = \int_0^1 \frac{\partial f}{\partial x_i}(tx_1, \dots, tx_n)dt.$$

Hence

$$\frac{\partial g_i}{\partial x_j}(x_1, \dots, x_n) = \int_0^1 \frac{\partial}{\partial x_j}\left(\frac{\partial f}{\partial x_i}(tx_1, \dots, tx_n)\right)dt = \int_0^1 \left(\sum_{k=1}^n \frac{\partial^2 f}{\partial x_i \partial x_k}(tx_1, \dots, tx_n)\frac{\partial(tx_k)}{\partial x_j}\right)dt =$$
$$= \int_0^1 t\frac{\partial^2 f}{\partial x_i \partial x_j}(tx_1, \dots, tx_n)dt.$$

Thus,

$$\frac{\partial g_i}{\partial x_j}(0) = \int_0^1 t\frac{\partial^2 f}{\partial x_i \partial x_j}(0)dt = \frac{\partial^2 f}{\partial x_i \partial x_j}(0)\int_0^1 tdt = \frac{1}{2}\frac{\partial^2 f}{\partial x_i \partial x_j}(0) \implies$$
$$\implies h_{ij}(0) = \frac{1}{2}\frac{\partial^2 f}{\partial x_i \partial x_j}(0)\int_0^1 dt = \frac{1}{2}\frac{\partial^2 f}{\partial x_i \partial x_j}(0)$$

Finally, we have f expressed similarly to a quadratic form. What we want to do next is diagonalize it, and to that end we will proceed by induction.

The base case of the induction ($r = 1$) is already proven.

Suppose now that there exists a coordinate system $(u_i, \dots, u_n)$ in a neighborhood $\mathcal{U}_1$ of 0, perhaps smaller than $\mathcal{U}$, such that

$$f = \pm(u_1)^2 \pm \cdots \pm (u_{r-1})^2 + \sum_{i,j \geq r} u_i u_j H_{ij}(u_1, \dots, u_n) \tag{1.1}$$

on all of $\mathcal{U}_1$ with $r > 1$ and $H_{ij} = H_{ji}$ for any i and j.

We can now perform a linear change of coordinates in $(u_r, \dots, u_n)$ to diagonalize $\sum_{i,j\geq r} u_i u_j H_{ij}(0)$.

We know that, the matrix (H_{ij}) being non-singular, the terms of the diagonal are all non-zero. Hence we may assume $H_{rr} \neq 0$. Let $g(u_1, \dots, u_n) = |H_{rr}(u_1, \dots, u_n)|^{1/2}$; it is clear that there exists a neighborhood $\mathcal{U}_2 \subset \mathcal{U}_1$ of 0 on which it is smooth and does not vanish. We define the new coordinate system

$$\begin{cases} V_i(u_1, \dots, u_n) = u_i, \quad i \neq r \\ V_r(u_1, \dots, u_n) = g(u_1, \dots, u_n)\left[u_r + \sum_{i>r} u_i \frac{H_{ir}(u_1,\dots,u_n)}{H_{rr}(u_1,\dots,u_n)}\right], \end{cases} \tag{1.2}$$

let us see that it is indeed one. We have that the Jacobian matrix of the coordinate transformation at 0 is

$$\frac{\partial(V_1, \dots, V_n)}{\partial(u_1, \dots, u_n)} = \begin{pmatrix} 1 & 0 & & \dots & & 0 \\ 0 & & & & & 0 \\ \vdots & & 1 & & & \vdots \\ \frac{\partial V_r}{\partial u_1} & \dots & & g(0) & \dots & \frac{\partial V_r}{\partial u_n} \\ 0 & & & & 1 & 0 \\ \vdots & & & & & \vdots \\ 0 & & & \dots & 0 & 1 \end{pmatrix},$$

and it is non-singular: it suffices to subtract from the r-th row the other rows multiplied by the appropriate term $\frac{\partial V_i}{\partial u_j}$ so that it becomes diagonal, and the terms of the diagonal are all non-zero.

It then follows from the *inverse function theorem* that the transformation is a diffeomorphism on a smaller neighborhood $\mathcal{U}_3 \subset \mathcal{U}_2$ of 0 and therefore $(v_1, \ldots, v_n)$ is a local coordinate system around p.

We now consider

$$u_r u_r H_{rr}(u_1, \ldots, u_n) + 2 \sum_{j=r+1}^{n} u_j u_r H_{jr}(u_1, \ldots, u_n). \tag{1.3}$$

Using the symmetry of H_{ij} we observe that this expression is nothing but the terms of (1.1) having $j = r$ or $i = r$. Comparing (1.1) and (1.2) we arrive at the fact that (1.3) is

$$\pm V_r V_r - \frac{1}{H_{rr}} \left[\sum_{i>r} u_i H_{ir}(u_1, \ldots, u_n) \right]^2 \tag{1.4}$$

Indeed,

$$\begin{aligned}
V_r V_r &= g^2 \left[u_r + \sum_{i>r} u_i \frac{H_{ir}}{H_{rr}} \right]^2 = \\
&= g^2 \left(u_r^2 + \left[\sum_{i>r} u_i \frac{H_{ir}}{H_{rr}} \right]^2 + 2u_r \sum_{i>r} u_i \frac{H_{ir}}{H_{rr}} \right) = \\
&= |H_{rr}| u_r^2 + \frac{|H_{rr}|}{H_{rr}^2} \left[\sum_{i>r} u_i H_{ir} \right]^2 + 2u_r \frac{|H_{rr}|}{H_{rr}} \sum_{i>r} u_i H_{ir} = \\
&= \pm H_{rr} u_r^2 \pm \frac{1}{H_{rr}} \left[\sum_{i>r} u_i H_{ir} \right]^2 \pm 2u_r \sum_{i>r} u_i H_{ir},
\end{aligned}$$

where the $\pm$ will be $+$ if H_{rr} is positive and $-$ if it is negative, and likewise in (1.4).

Thus, it is clear that (1.3) and (1.4) are equivalent. And from here it is easy to see how (1.1) turns into

$$f = \pm (V_1)^2 \pm \cdots \pm (V_{r-1})^2 + (V_r)^2 \pm \sum_{i,j>r} V_i V_j \tilde{H}_{ij}(V_1, \ldots, V_n)$$

for some new symmetric terms $\tilde{H}_{ij}$ in terms of the new coordinate system.

This concludes the induction, since we have one more quadratic summand, and the existence of the sought local coordinate system is proven.

$\square$

Remark 1.2.9. A part of the process remains implicit throughout the proof. When defining the new coordinate system in the induction step, what we are doing is defining a function we may call $l : \mathbb{R}^n \to \mathbb{R}^n$

that, given the coordinates $(u_1(p), \ldots, u_n(p))$ of a point $p \in M$, takes them to new ones $(v_1(p), \ldots, v_n(p))$ and is a diffeomorphism between open sets of $\mathbb{R}^n$. Thus, if the chart inducing the coordinate system $(u_1, \ldots, u_n)$ was $\varphi : M \to \mathbb{R}^n$, the one inducing $(v_1, \ldots, v_n)$ will be $l \circ \varphi$.

The following corollary will be key when proving one of the main theorems of Morse theory (see 2.2.17) and shows one of the fundamental advantages of working with non-degenerate critical points.

Corollary 1.2.10. *Let M be a smooth manifold. Every non-degenerate critical point of a smooth function from M to $\mathbb{R}$ is isolated from the other critical points.*

Proof. Let $p \in M$ be a non-degenerate critical point of a function f. There exists a local coordinate system $(y_1, \ldots, y_n)$ in a neighborhood $\mathcal{U}$ of p such that $y_i(p) = 0$ for all $i \in \{1, \ldots, n\}$ and

$$f = f(p) - (y_i)^2 - \cdots - (y_\lambda)^2 + (y_{\lambda+1})^2 + \cdots + (y_n)^2$$

by the Morse Lemma (1.2.8).

Suppose there exists another critical point $q \in \mathcal{U}$, then

$$\frac{\partial f}{\partial x_i}(q) = \pm 2q_i = 0 \iff q_i = y_i(q) = 0 \iff q_i = p_i$$

Hence p is the only critical point of f in $\mathcal{U}$.

□

Next we present some degenerate and non-degenerate critical points together with the graphs of their corresponding functions, Figures 1.1, 1.2 and 1.3. Observe that while non-degenerate points must be isolated, a function can have degenerate critical points that are either isolated or non-isolated.

Example 1.2.11.

1. The function $f(x, y) = (x^2 - 1)^2 + (y^2 - 1)^2$ has nine non-degenerate critical points. It is easy to check that $\text{Hess}_p f$ is a diagonal matrix whose entries do not vanish at any of the critical points.

2. The function $g(x, y) = x^3 - 3xy^2$, whose graph corresponds to the “monkey saddle”, possesses a single degenerate critical point at the point $(0, 0)$. It suffices to see that $\text{Hess}_{(0,0)} f$ has zeros in all four entries.

3. We observe that $h(x, y) = x^2 y^2$ has infinitely many degenerate critical points, those of the form $(0, y)$ and $(x, 0)$.

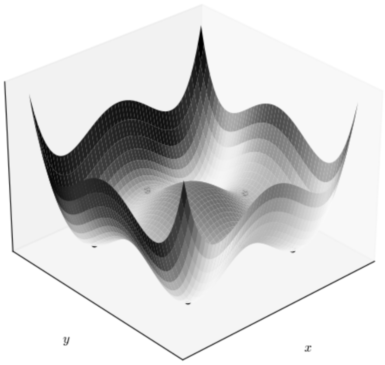

Figure 1.1: $f(x,y) = (x^2-1)^2+(y^2-1)^2$

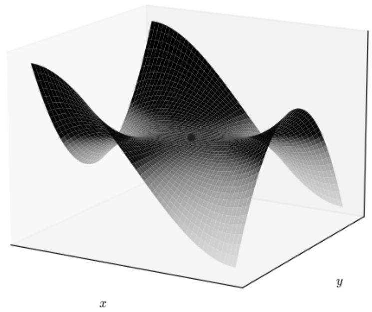

Figure 1.2: $g(x,y) = x^3 - 3xy^2$

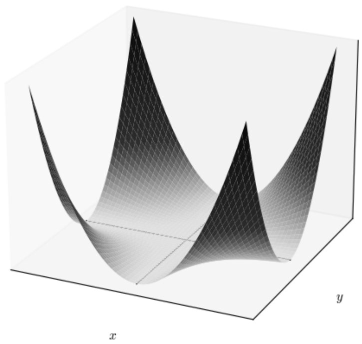

Figure 1.3: $h(x,y) = x^2y^2$

We conclude the chapter by defining the objects that give it its name and of which we will make use from now on to describe the CW-complex structure of any smooth manifold.

Definition 1.2.12. We will say that a smooth function $f : M \to \mathbb{R}$ is a *Morse function* if all its critical points are non-degenerate.

Chapter 2

The CW-complex structure of a manifold

The objective of this chapter is to prove that every manifold admitting a Morse function whose sublevel sets are compact has the homotopy type of a CW-complex (2.2.17). To that end, we formally introduce what we understand by CW-complex and by homotopy type.

We also include two results necessary for our objective, but interesting in their own right, which describe under what conditions and how the homotopy type changes as we traverse the sublevel sets of a Morse function (2.2.5 and 2.2.7).

The chapter concludes with two examples of how the main theorems proven can be applied.

2.1 CW-complexes and homotopy type

We will say that two continuous functions $f, g : X \to Y$ between two topological spaces are *homotopic* if there exists a *homotopy* between them, that is, if there exists a continuous function $H : X \times [0,1] \to Y$ satisfying

$$H(x,0) = f(x) \quad \text{and} \quad H(x,1) = g(x)$$

for all $x \in X$.

The homotopy relation is an equivalence relation. We will denote that two functions are related by means of the symbol $\simeq$.

When denoting or constructing homotopies we will use two notations interchangeably. We refer to a homotopy between two continuous functions f and g as the continuous function $H(x,t) : X \times [0,1] \to Y$ satisfying the above conditions, or as the family of continuous functions $\{H_t : X \to Y\}_{t\in[0,1]}$ such that $H_t(x) = H(x,t)$ for any t and x.

Definition 2.1.1. Let X and Y be topological spaces. We say that X and Y are *homotopy equivalent* if there exist continuous maps

$$f : X \to Y, \qquad g : Y \to X$$

such that

$$g \circ f \simeq \mathrm{id}_X \qquad \text{and} \qquad f \circ g \simeq \mathrm{id}_Y.$$

In this case, we say that X and Y have the same *homotopy type* and we call f and g *homotopy equivalences.*

A *retraction* of a space X onto a subspace $A \subseteq X$ is a continuous map $r : X \to A$ satisfying $r|_A = \mathrm{id}_A$; in that case, we call A a *retract* of X. Moreover, a homotopy $\{H_t\}$ is said to be *relative to* A if $H_t|_A = \mathrm{id}_A$ for all $t \in [0,1]$.

Definition 2.1.2. Let X be a topological space and $A \subset X$ a subspace. We say that A is a *strong deformation retract* of X if there exists a homotopy relative to A from the identity function of X, id_X, to a retraction $r : X \to A$.

Note. In what follows we will refer to strong deformation retracts simply as *deformation retracts.*

Remark 2.1.3. Note that every deformation retract of a space has the same homotopy type as the latter. Given a space X and a deformation retract $A \subseteq X$, the inclusion function, $i : A \hookrightarrow X$, and the retraction $r : X \to A$ are homotopy equivalences.

In what follows we will denote by D^n the closed unit disk, whose boundary is $\partial D^n = \mathbb{S}^{n-1}$, the $(n-1)$-dimensional unit sphere.

Definition 2.1.4. A *cell* e^n is a topological space homeomorphic to the interior of the n-dimensional unit disk D^n. We will say that the dimension of the cell is n and that e^n is an *n-cell.*

Definition 2.1.5. We call *CW-complex* a topological space constructed following these steps:

(i) One starts with a discrete set X_0 (also called the *0-skeleton*), whose elements we will call *0-cells.*

(ii) One constructs, inductively, the so-called n-skeleton for $n > 0$ as

$$X_n = X_{n-1} \cup_{\varphi_\alpha} D^n_\alpha, \; \alpha \in \Lambda_n.$$

That is, X_n is obtained from X_{n-1} by attaching to it a collection, indexed by Λ_n, of n-cells by means of the *attaching maps* $\varphi_\alpha : \partial D^n_\alpha \to X_{n-1}$.

(iii) We will call *CW-complex* the space

$$X = \bigcup_n X_n.$$

Remark 2.1.6.

1. Note that the attaching maps are only required to be continuous.
2. We say that the dimension of a CW-complex X is m if the construction has a finite number m of steps; in that case, $X = X_m$. If the number of steps is infinite, the CW-complex will have infinite dimension.

3. More formally, we call an n-cell the image of the inclusion $i_\alpha : D^n_\alpha \to X_n$ and we endow X with the *weak topology*, that is, $\mathcal{U} \subseteq X$ is open if and only if $i_\alpha^{-1}(\mathcal{U}) \subseteq D^k_\alpha$ is open for all α and $k \geq 0$. In fact, it is this topology that gives them their name, since CW comes from the English "Closure finite-Weak topology"

CW-complexes present a convenient structure for applying algebraic techniques and computing topological invariants such as the fundamental group or the homology groups, which we will see later on. There is also the advantage that every smooth manifold can be endowed with a structure of this kind either via triangulations (see Section 3.1.1), or via Morse theory.

Note. We will habitually call *vertices* the 0-cells, *edges* the 1-cells and *faces* the 2-cells. Note also that the boundary or frontier of an edge is two vertices, and that of a face a finite collection of edges.

We refer the reader to Section 3.2 on cellular homology, in which we develop some examples of CW-complexes.

Definition 2.1.7. To each cell e^n_α of a CW-complex X there corresponds a *characteristic map* $\phi_\alpha : D^n_\alpha \to X$ which can be factored as

$$D^n_\alpha \hookrightarrow X_{n-1} \sqcup_\alpha D^n_\alpha \to X_n \hookrightarrow X$$

where the middle function is the quotient map defining X_n (that is, the one identifying the points of D^n_α with their images under φ_α) and the others the natural inclusions. ϕ_α extends the attaching map φ_α and is a homeomorphism from the interior of D^n_α onto e^n_α.

Proposition 2.1.8. *Given the n-skeleton X_n and the $(n-1)$-skeleton X_{n-1} of a CW-complex X we have that*

$$X_n/X_{n-1} = \bigvee_{\alpha\in\Lambda_n} D^n_\alpha/\partial D^n_\alpha = \bigvee_{\alpha\in\Lambda_n} \mathbb{S}^{n-1}_\alpha,$$

where $\vee$ denotes the wedge sum [Hat02, p. 10] and Λ_n runs over the n-cells of X.

This last fact follows from the fact that each n-cell e^n_α is obtained by attaching the boundary of an n-disk to the $(n-1)$-skeleton. When taking the quotient X_n/X_{n-1} we are collapsing the $(n-1)$-skeleton to a single point, so that composing the attaching map with the quotient map what one obtains is the trivial class. Thus, the boundary of the disk becomes identified to a single point; moreover, we know that $D^n/\partial D^n \cong \mathbb{S}^n$.

Since the boundaries of all the n-disks become identified to the same point, what one obtains is a "bouquet" of spheres, which we denote

$$\bigvee_{\alpha\in\Lambda_n} \mathbb{S}^{n-1}_\alpha.$$

2.2 On the attaching of λ-cells

In this section we will lay the foundations of Morse Theory. Up to this point we have provided the technical apparatus upon which to build the theory; here we will see how Morse functions can serve to relate the homotopy type of a manifold with that of a CW-complex with as many λ-cells as critical points of index λ f has.

We begin by introducing some results on vector fields (respecting the vocabulary of [Mil63]) that will serve us to prove the main results of the section.

Definition 2.2.1. Let M be a smooth manifold. We call a *one-parameter group of diffeomorphisms of* M a map $\varphi : \mathbb{R} \times M \to M$ satisfying:

i) For every $t \in \mathbb{R}$, $\varphi_t : M \to M$ defined as $\varphi_t(q) = \varphi(t, q)$, is a diffeomorphism of M onto itself.

ii) For all $t, s \in \mathbb{R}$ one has $\varphi_{t+s} = \varphi_t \circ \varphi_s$.

Given a one-parameter group of diffeomorphisms φ we define the following vector field on M

$$X_q(f) = \lim_{h \to 0} \frac{f(\varphi_h(q)) - f(q)}{h},$$

where f is a smooth function from M to $\mathbb{R}$, and we say that X generates φ.

Normally, one refers to the curve $\varphi_\bullet(q) : \mathbb{R} \to M$, $t \mapsto \varphi_t(q)$ as the *flow of* X when the vector field X generating the one-parameter group of diffeomorphisms is known. In any case, we will keep a vocabulary as faithful as possible to the text [Mil63].

We will also speak of the *velocity vector* of a differentiable curve $c : \mathbb{R} \to M$, which we define similarly to X,

$$\frac{dc}{dt}(f) = \lim_{h \to 0} \frac{f(c(t+h)) - f(c(t))}{h} \in T_{c(t)}M.$$

Lemma 2.2.2. *Every vector field on a smooth manifold M that vanishes outside a compact set $K \subset M$ generates a unique one-parameter group of diffeomorphisms of M.*

Proof. Let X be a vector field and φ a one-parameter group of diffeomorphisms of M generated by X. Then we have that, fixing any point $q \in M$, the curve $\varphi_\bullet(q) : \mathbb{R} \to M$, $t \mapsto \varphi_t(q)$ satisfies the differential equation

$$\frac{d\varphi_t}{dt}(q) = X_{\varphi_t(q)} \tag{$*$}$$

with initial condition $\varphi_0(q) = q$.

Indeed,

$$\frac{d\varphi_t(q)}{dt}(f) = \lim_{h \to 0} \frac{f(\varphi_{t+h}(q)) - f(\varphi_t(q))}{h} = \lim_{h \to 0} \frac{f(\varphi_h(\varphi_t(q))) - f(\varphi_t(q))}{h} = \lim_{h \to 0} \frac{f(\varphi_h(p)) - f(p)}{h} = X_p(f),$$

with $p = \varphi_t(q)$.

Now, by the Picard–Lindelöf theorem we know that a unique solution exists locally, depending differentiably on the initial condition. That is, for every $p \in M$ there exist a neighborhood $\mathcal{U}$ of p and $\varepsilon > 0$ such that $(*)$ has a unique solution for all $q \in \mathcal{U}$ and $|t| < \varepsilon$.

Since $K \subset M$ is compact, it can be covered by a finite number of such neighborhoods. Let ε_0 be the smallest of the values defined above and let us set $\varphi_t(q) = q, \quad \forall q \notin K$ (hence $X_q = 0$ outside K). Then $(*)$ has a unique solution $\varphi_t(q)$ for $|t| < \varepsilon_0$ and for all $q \in M$, smooth with respect to both variables.

Let us see that for every suitable t each φ_t is a diffeomorphism of M onto itself. For this it suffices to check that $\varphi_{t+s} = \varphi_t \circ \varphi_s$ holds, assuming $|t|, |s|, |t+s| < \varepsilon_0$.

Let $q \in M$ be fixed and $p = \varphi_s(q)$, then $\varphi_t(p) = \varphi_t(\varphi_s(q))$. We define the curve $\gamma(u) = \varphi_{u+s}(q)$; thus, $\gamma(0) = p$ and

$$\frac{d\gamma}{du} = \frac{d\varphi_{u+s}}{du}(q) = X_{\varphi_{u+s}(q)} = X_{\gamma(u)}.$$

Hence γ is also a solution of $(*)$ with initial condition $\gamma(0) = p$.

However, $\varphi_u(p)$ is also a solution with initial condition $\varphi_0(p) = p$. Hence by uniqueness of solutions

$$\varphi_{u+s}(q) = \gamma(u) = \varphi_u(p) = \varphi_u(\varphi_s(q)) = \varphi_u \circ \varphi_s(q),$$

whenever $|u|, |u+s| < \varepsilon_0$.

Lastly, we need to define φ_t for all $t \in \mathbb{R}$.

Let $t \in \mathbb{R}$. Let $k \in \mathbb{Z}$ and $r \in \mathbb{R}$ with $|r| \leq \frac{\varepsilon_0}{2}$ be the unique values such that $t = k\frac{\varepsilon_0}{2} + r$. Then we define

$$\varphi_t = \underbrace{\varphi_{\frac{\varepsilon_0}{2}} \circ \cdots \circ \varphi_{\frac{\varepsilon_0}{2}}}_{k\ times} \circ \varphi_r.$$

It is clear that $\varphi : \mathbb{R} \times M \to M$ is a unique diffeomorphism for every $t \in \mathbb{R}$.

$\square$

Definition 2.2.3. Let $(M, \langle \cdot, \cdot \rangle)$ be a Riemannian manifold and $f : M \to \mathbb{R}$ smooth. The *gradient of* f is defined as the vector field ∇f satisfying

$$\langle X, \nabla f \rangle = X(f)$$

for any other vector field X.

Definition 2.2.4. Let M be a smooth manifold and $f : M \to \mathbb{R}$ smooth. We define the *sublevel sets of* f as

$$M^a := f^{-1}(-\infty, a] = \{p \in M : f(p) \leq a\}.$$

Theorem 2.2.5. *Let $f : M \to \mathbb{R}$ be smooth on a Riemannian manifold $(M, \langle\cdot,\cdot\rangle)$. Let $a < b$ be real numbers such that $f^{-1}[a,b]$ is compact and contains no critical point of f. Then M^a is diffeomorphic to M^b. Moreover, M^a is a deformation retract of M^b, hence they have the same homotopy type.*

Proof. Let $c : \mathbb{R} \to M$ be a curve with velocity vector $\frac{dc}{dt}$, then

$$\langle \frac{dc}{dt}, \nabla f\rangle = \frac{dc}{dt}(f) = \frac{d(f\circ c)}{dt}.$$

We define the smooth map $\rho : M \to \mathbb{R}$ as

$$\rho(p) := \begin{cases} \frac{1}{\langle \nabla_p f, \nabla_p f\rangle} & , \ p \in f^{-1}[a,b] \\ 0 & , p \notin K, \end{cases}$$

where K is a compact neighborhood of $f^{-1}[a,b]$

Now, let the vector field be given by $X_q = \rho(q)\nabla_p f$. We have that X_p vanishes outside K, hence by Lemma 2.2.2 it generates a unique one-parameter group of diffeomorphisms φ_t of M.

Let us fix $q \in M$ and consider $f \circ \varphi_\bullet(q)$, $t \mapsto f(\varphi_t(q))$. If $\varphi_t(q)$ is in $f^{-1}[a,b]$ we have

$$\begin{aligned}\frac{df(\varphi_t(q))}{dt} &= \left\langle \frac{d\varphi_t(q)}{dt}, \nabla f\right\rangle = \langle X, \nabla f\rangle = \langle \rho\ \nabla f, \nabla f\rangle = \\ &= \left\langle \frac{1}{\langle\nabla f, \nabla f\rangle}\nabla f, \nabla f\right\rangle = \frac{\langle\nabla f, \nabla f\rangle}{\langle\nabla f, \nabla f\rangle} = 1,\end{aligned}$$

so $f(\varphi_t(q))$ is linear with derivative 1 as long as $f(\varphi_t(q))$ is in $[a,b]$, that is, $f(\varphi_t(q)) = f(q) + t$.

Let us now see that φ_{b-a} is a diffeomorphism from M^a to M^b.

Let $q \in M$,

$$\varphi_{b-a}(q) \in M^b \iff (\varphi_{b-a}(q)) = f(q) + b - a \leq b \iff f(q) - a \leq 0 \iff q \in M^a,$$

and in particular it sends the boundary of M^a to the boundary of M^b

$$q \in \partial M^a \iff f(q) = a \implies f(\varphi_{b-a}(q)) = b \iff b \in \partial M^b.$$

To conclude the proof it remains to check that M^a is indeed a deformation retract of M^b. For this we are going to construct a homotopy relative to M^a from the identity of M^b to a retraction of M^b onto M^a.

Let $H(t,q) : [0,1] \times M^b \to M^b$ be defined as

$$H(t,q) = \begin{cases} q, & f(q) \leq a \iff q \in M^a, \\ \varphi_{t(a-f(q))}(q), & a \leq f(q) \leq b. \end{cases}$$

Then, $H(0,\cdot) = id_{M^b}$ since $\varphi_0(q) = q$, $\forall q \in M^b$, and $H(1,\cdot)$ is a retraction of M^b onto M^a; let us check it. Let $q \in f^{-1}[a,b]$, we have

$$H(1,q) = f(\varphi_{(a-f(q))}(q)) = f(q) + (a - f(q)) = a \iff \varphi_{(a-f(q))}(q) \in f^{-1}(a),$$

hence $H(1, H(1,\cdot)) = id_{M^a}$.

$\square$

Remark 2.2.6. Note the importance of $f^{-1}[a,b]$ being compact. Let $a = -1/2$ and $b = 1/2$ and let our manifold be the sphere $\mathbb{S}^2$ embedded in $\mathbb{R}^3$; it is clear that the function assigning to each point of the sphere its z coordinate is Morse and satisfies that $f^{-1}[a,b]$ is compact since it is closed and the sphere is compact. Now, if we remove from $\mathbb{S}^2$ a point p such that $f(p)$ lies between $f(a)$ and $f(b)$, we have that $f^{-1}[a,b]$ ceases to be compact (as it ceases to be closed and we are in a Hausdorff space) and clearly M^a is not a deformation retract of M^b. Indeed, M^a is homeomorphic to a disk and M^b to a punctured disk, which is not simply connected, hence they do not have the same homotopy type.

The following theorem is crucial for knowing the CW-complex structure of a manifold. It establishes that the change produced in the homotopy type of the sublevel sets of a Morse function upon crossing a critical point is equivalent to that of attaching a cell of dimension the index of said critical point.

Theorem 2.2.7. *Let $f : M \to \mathbb{R}$ be smooth on a Riemannian manifold $(M, \langle\cdot,\cdot\rangle)$ and $p \in M$ a non-degenerate critical point of f with index λ. Suppose that $f(p) = c$ and that there exists $\varepsilon > 0$ such that $f^{-1}[c-\varepsilon, c+\varepsilon]$ is compact and contains no critical point of f other than p. Then, for every sufficiently small $\varepsilon > 0$, $M_f^{c+\varepsilon}$ has the homotopy type of $M_f^{c-\varepsilon}$ with a λ-cell attached.*

Proof. We are going to define a function F that is smaller than f in a neighborhood $\mathcal{U}$ of p, with the aim of leaving p outside $F^{-1}[c-\varepsilon, c+\varepsilon]$ and being able to apply Theorem 2.2.5 to F. In this way, we will have proven that $F^{-1}(-\infty, c-\varepsilon] = M_F^{c-\varepsilon}$ is a deformation retract of $M_f^{c+\varepsilon}$.

Afterwards, we will prove that $M_f^{c-\varepsilon}$ with a λ-cell attached is a deformation retract of $M_F^{c-\varepsilon}$ and by the transitivity of this relation we will be done.

Let $(\mathcal{U}, \phi)$ be a chart of M containing p and $(u^1, \ldots, u^n)$ the induced coordinates such that $u^1(p) = \cdots = u^n(p) = 0$. In order to simplify the expressions that follow we define the functions $\xi : \mathcal{U} \to [0, \infty)$ and $\eta : \mathcal{U} \to [0, \infty)$ as

$$\xi(q) = (u^1(q))^2 + \cdots + (u^\lambda(q))^2$$
$$\eta(q) = (u^{\lambda+1}(q))^2 + \cdots + (u^n(q))^2$$

Applying the Morse Lemma (1.2.8) to f we obtain

$$f(q) = c - \xi(q) + \eta(q)$$

for each $q \in \mathcal{U}$, where $c = f(p)$.

Let $\varepsilon > 0$ be small enough so that:

i) $f^{-1}[c-\varepsilon, c+\varepsilon]$ is compact and contains no critical point of f other than p.

ii) $\phi(\mathcal{U}) \subset \mathbb{R}^n$ contains the closed ball

$$\phi\left(\overline{B(0,\sqrt{2\varepsilon})}\right) = \{q \in \mathcal{U} : \xi(q) + \eta(q) \leq 2\varepsilon\}$$

Let us construct $F : M \to \mathbb{R}$. Let $\mu : \mathbb{R} \to \mathbb{R}$ be a smooth map satisfying the following

$$\begin{cases} \mu(0) > \varepsilon \\ \mu(r) = 0, \quad r \geq 2\varepsilon \\ -1 < \mu'(r) \leq 0, \quad \forall r \in \mathbb{R}, \end{cases}$$

then we define F as

$$F(q) = \begin{cases} f(q), & q \notin \mathcal{U}, \\ f(q) - \mu(\xi(q) + 2\eta(q)), & q \in \mathcal{U}. \end{cases}$$

It is clear that F is well defined since μ vanishes on $\partial\mathcal{U}$. Moreover, it is smooth since in a neighborhood of $\partial\mathcal{U}$ one has that μ' is identically 0, hence both approaching from inside $\mathcal{U}$ and from outside $d_qF = d_qf$.

Now that we have F we must check that we are within the hypotheses of Theorem 2.2.5. Let us see first that $M_F^{c+\varepsilon} = M_f^{c+\varepsilon}$, then that F and f possess the same critical points and that $F^{-1}[c-\varepsilon, c+\varepsilon]$ contains no critical point.

Indeed, $\mu(\xi(q) + 2\eta(q)) = 0$ for every $q \in M$ such that $\xi(q) + 2\eta(q) \geq 2\varepsilon$, hence outside the ellipsoid $E = \{q \in M : \xi(q) + 2\eta(q) \leq 2\varepsilon\}$ we have that $F = f$.

We know that $E \subset M_f^{c+\varepsilon}$ and by

$$F(q) < f(q) = c - \xi(q) + \eta(q) \leq c + \frac{1}{2}\xi(q) + \eta \leq c + \varepsilon, \ \forall q \in E,$$

we have that $E \subset M_F^{c+\varepsilon}$. Seeing that it is the only region on which they differ, we conclude that $M_F^{c+\varepsilon} = M_f^{c+\varepsilon}$.

Now let us see that their critical points coincide. We observe that on $\mathcal{U}$ the following hold

$$\frac{\partial F}{\partial \xi} = -1 - \mu'(\xi + 2\eta) < 0 \quad \text{and}$$
$$\frac{\partial F}{\partial \eta} = 1 - 2\mu'(\xi + 2\eta) \geq 1.$$

The critical points of F will be those at which its partial derivatives vanish

$$\frac{\partial F}{\partial x_i} = \frac{\partial F}{\partial \xi}\frac{\partial \xi}{\partial x_i} + \frac{\partial F}{\partial \eta}\frac{\partial \eta}{\partial x_i},$$

but the only point simultaneously satisfying $\frac{\partial \xi}{\partial x_i} = 0$ and $\frac{\partial \eta}{\partial x_i} = 0$ is p, since otherwise f would have more critical points in $\mathcal{U}$. Hence, as they coincide outside $\mathcal{U}$, we have that F and f have the same critical points.

Let us see that $F^{-1}[c-\varepsilon, c+\varepsilon] = M_F^{c+\varepsilon} \setminus M_F^{c-\varepsilon}$ is compact and contains no critical point. It is clear that $M_f^{c-\varepsilon} \subset M_F^{c-\varepsilon}$ since F and f coincide except on the neighborhood of p on which $F \leq f$, thus

$$M_F^{c+\varepsilon} = M_f^{c+\varepsilon} \implies M_F^{c+\varepsilon} \setminus M_F^{c-\varepsilon} \subset M_f^{c+\varepsilon} \setminus M_f^{c-\varepsilon} = f^{-1}[c-\varepsilon, c+\varepsilon],$$

whereby $F^{-1}[c-\varepsilon, c+\varepsilon]$ is compact. On the other hand,

$$F(p) = c - \mu(0) < c - \varepsilon,$$

hence $p \notin F^{-1}[c-\varepsilon, c+\varepsilon]$.

In this way, we find ourselves within the hypotheses of Theorem 2.2.5 for F, which proves that $M_F^{c-\varepsilon}$ is a deformation retract of $M_F^{c+\varepsilon} = M_f^{c+\varepsilon}$.

Let $e^\lambda = \{q \in \mathcal{U} : \xi(q) \leq \varepsilon,\ \eta(q) = 0\} \subset \mathcal{U}$ be a λ-cell. First of all, let us see that

$$\partial e^\lambda = \{q \in \mathcal{U} : \xi(q) = \varepsilon, \quad \eta(q) = 0\} = M_f^{c-\varepsilon} \cap e^\lambda.$$

Indeed:

($\subseteq$) For all $q \in \partial e^\lambda,\ f(q) = c - \varepsilon \implies q \in M_f^{c-\varepsilon}$.

($\supseteq$) For all $q \in M_f^{c-\varepsilon} \cap e^\lambda$, we have $\xi(q) \leq \varepsilon,\ \eta(q) = 0$ and $f(q) \leq -\varepsilon$, then since

$$f(q) = c - \xi(q) \leq c - \varepsilon \iff \xi(q) \geq \varepsilon$$

necessarily $\xi(q) = \varepsilon$, hence $q \in \partial e^\lambda$.

Figure 2.1 shows the sublevel sets of f and the λ-cell. The axes correspond to the planes $u^{\lambda+1} = \cdots = u^n = 0$ and $u^1 = \cdots = u^\lambda = 0$. The circle represents the boundary of the ball of radius $\sqrt{2\varepsilon}$ and center p. The darkest region corresponds to $M_f^{c-\varepsilon}$, the region with large dots is $f^{-1}[c-\varepsilon, c]$ and, finally, the region with the smallest dots is $f^{-1}[c, c+\varepsilon]$.

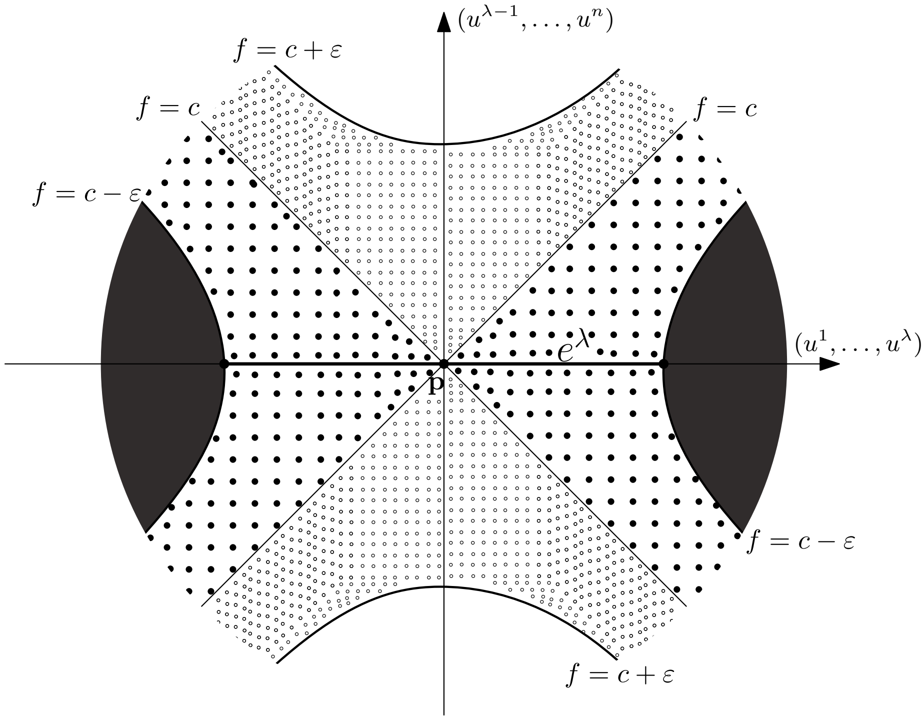

Figure 2.1: Schematic representation of the neighborhood of the critical point p with respect to f.

For the rest of the proof it will turn out convenient to introduce the notion of *handle*. We will understand $M_F^{c-\varepsilon}$ as $M_f^{c-\varepsilon} \cup H$ where $H = \overline{M_F^{c-\varepsilon} \setminus M_f^{c-\varepsilon}}$ is the handle we speak of, denoted this way by virtue of its English name *handle*.

We observe that e^λ is contained in H. Indeed, F attains its maximum on $\mathcal{U}$ when $\mu(\xi + 2\eta)$ attains its minimum, and the latter attains it when ξ and η vanish, that is, only at p, thus

$$F(q) \leq F(p) < c - \varepsilon \ \text{ and } \ f(q) = c - \xi(q) \geq c - \varepsilon, \ \forall q \in e^\lambda,$$

hence $q \in e^\lambda$ which implies that q is in H.

Now we want to prove that $M_f^{c-\varepsilon} \cup e^\lambda$ is a deformation retract of $M_F^{c-\varepsilon} = M_f^{c-\varepsilon} \cup H$. For this we are going to construct a homotopy relative to $M_f^{c-\varepsilon} \cup e^\lambda$ from the identity to a retraction onto $M_f^{c-\varepsilon} \cup e^\lambda$, as in the proof of Theorem 2.2.5.

Figure 2.2 shows the neighborhood of p from this perspective. The dark zone represents $M_f^{c-\varepsilon}$, the dotted zone corresponds to $F^{-1}[c-\varepsilon, c+\varepsilon] = M_F^{c+\varepsilon} \setminus M_F^{c-\varepsilon}$ and the zone with arrows is H, the handle. Besides identifying H, the arrows illustrate the deformation we will carry out in the next and final step of the proof.

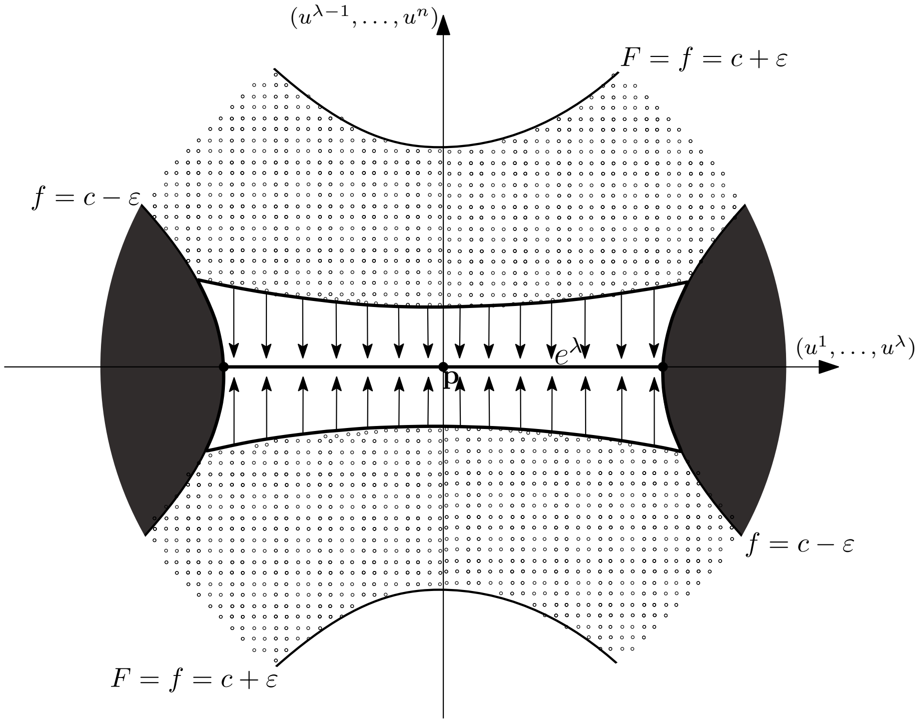

Figure 2.2: Schematic representation of the neighborhood of the critical point p with respect to F and with the handle.

Let $\Psi:[0,1]\times M_f^{c-\varepsilon}\cup H\to M_f^{c-\varepsilon}\cup H$; we define Ψ as the identity outside $\mathcal{U}$. On $\mathcal{U}$ we must distinguish three regions.

Figure 2.3 shows schematically the regions into which $\mathcal{U}$ is divided near the handle. The zone with black-tipped arrows represents the part of region 1 lying inside the handle. The zone with the small hollow-tipped arrows coincides with the part of region 2 lying inside the handle. The dark zone is region 3. Likewise, the arrows illustrate which part of the handle we want to deform and onto what. Note that in region 3 nothing needs to be deformed.

Region 1: $R_1=\{q\in\mathcal{U}:\xi(q)\le\varepsilon\}$.

In this region what we want is to take the points to e^λ.

We define $\Psi(t,q)=\Psi(t,(u_1(q),\ldots,u_n(q))=(u_1(q),\ldots,u_\lambda(q),tu_{\lambda+1}(q),\ldots,tu_n(q))$.

It is clear that it is relative to $M_f^{c-\varepsilon}\cup e^\lambda$ ($R_1\cap M_f^{c-\varepsilon}=\varnothing$ and it does not touch the points of e^λ) and that $\Psi(1,\cdot)$ is the identity. Moreover,

$$\forall q\in R_1,\ \Psi(0,q)\in e^\lambda \ \text{ and } \ \Psi(0,\Psi(0,q))=\Psi(0,q).$$

Likewise, since $\frac{\partial F}{\partial \eta} > 0$ (that is, F grows in the direction of η) we have that $F(\Psi(t,q)) \leq F(q)$ for all $t \in [0,1]$ and all $q \in R_1$. Hence Ψ takes $M_f^{c-\varepsilon} \cup H$ into itself.

Region 2: $R_2 = \{q \in \mathcal{U} : \ \varepsilon \leq \xi(q) \leq \eta(q) + \varepsilon\}$.

Here what we seek is to take the points of R_2 to the boundary of $M_f^{c-\varepsilon}$.

We define $\Psi(t,q) = \Psi(t,(u_1(q),\ldots,u_n(q)) = (u_1(q),\ldots,u_\lambda(q), s_t(q)u_{\lambda+1}(q),\ldots,s_t(q)u_n(q))$, where

$$s_t(q) = t + (1-t)\left(\frac{\xi(q)-\varepsilon}{\eta}\right)^{1/2} \in \ [0,1]$$

Just as on R_1 it is clear that it is relative to $M_f^{c-\varepsilon} \cup e^\lambda$, that $D(1,\cdot)$ is again the identity and that $\Psi(0,\cdot)$ is a retraction onto the boundary of $M_f^{c-\varepsilon}$. The latter follows from the fact that on the boundary of $M_f^{c-\varepsilon}$ one has $-\xi + \eta = \varepsilon$, hence $s_0(q) = 1$.

It remains to see that s_t is continuous as $\xi \to \varepsilon$ and $\eta \to 0$. Rearranging the inequality defining R_2 we have $0 \leq \xi - \varepsilon \leq \eta$, which means that as η tends to zero, ξ must tend to ε faster. Thus,

$$0 \leq \lim_{\xi \to \varepsilon, \eta \to 0} \frac{\xi - \varepsilon}{\eta} \leq 1$$

and in particular, we have that $\lim_{\xi \to \varepsilon, \eta \to 0} \frac{\xi-\varepsilon}{\eta} = 0$ necessarily. We conclude then that s_t generates no problems in the continuity of Ψ near the points with $\eta(q) = 0$.

Region 3: $R_3 = \{q \in \mathcal{U} : \ \eta(q) + \varepsilon \leq \xi(q)\}$

R_3 coincides with $M_f^{c-\varepsilon}$, hence it suffices to take $\Psi(t,q) = Id_{M_f^{c-\varepsilon}}$.

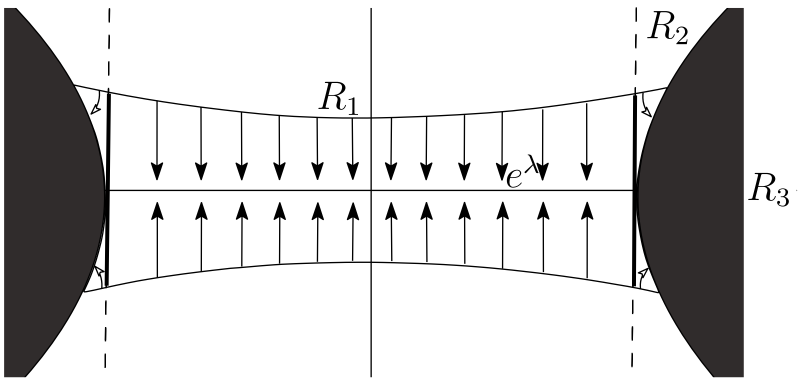

Figure 2.3: Schematic representation of the regions into which the handle is divided.

The three regions completely cover $\mathcal{U}$. This concludes the proof that $M_f^{c-\varepsilon} \cup e^\lambda$ is a deformation retract of $M_F^{c-\varepsilon} = M_f^{c-\varepsilon} \cup H$ and with it the proof of the theorem.

$\square$

Remark 2.2.8.

(i) By modifying the proof a little it is possible to prove that M_f^c is also a deformation retract of $M_f^{c+\varepsilon}$. In fact, M_f^c is a deformation retract of M_F^c, which is in turn a deformation retract of $M_f^{c+\varepsilon}$; this, together with the theorem, makes it clear that $M_f^{c-\varepsilon} \cup e^\lambda$ is a deformation retract of M_f^c.

(ii) Furthermore, as a corollary of this proof it follows that $M_f^{c+\varepsilon}$ is not only homotopy equivalent but homeomorphic to $M^{c-\varepsilon}$ with a λ-handle $D^\lambda \times D^{n-\lambda}$ attached. In case M is compact, doing this at each critical point gives rise to a decomposition of M into successive attachments of handles known as a "handlebody decomposition". This decomposition is the one used, for example, in the proof of the h-cobordism theorem.

Next we state the general case of this result, whose proof is similar.

Theorem 2.2.9. *Let f and M be within the hypotheses of the previous theorem. Let $p_1, \dots, p_k$ be non-degenerate critical points of f with indices $\lambda_1, \dots, \lambda_k$ such that $f(p_i) = c$, $\forall i \in \{1, \dots, k\}$. Then there exists a sufficiently small $\varepsilon > 0$ such that $M_f^{c+\varepsilon}$ has the homotopy type of*

$$M_f^{c-\varepsilon} \cup e^{\lambda_1} \cup \cdots \cup e^{\lambda_k}.$$

The last result crowns the section by relating the homotopy type of a smooth manifold with that of a CW-complex whose cells are defined by the critical points of a Morse function.

Its proof will require the *cellular approximation theorem*, whose proof can be found in [Hat02, p. 349], Theorem 1 of [Whi49a], and two lemmas about a topological space with a cell attached which we will prove. We proceed in this order.

Definition 2.2.10. A function $f : X \to Y$ between CW-complexes is said to be a *cellular map* if $f(X_k) \subset Y_k$ for every k. That is, a function taking cells to cells of the same or smaller dimension.

Theorem 2.2.11 (Cellular approximation theorem)**.** *Every function $f : X \to Y$ between CW-complexes is homotopic to a cellular map.*

Definition 2.2.12. A topological space X is said to *dominate* a topological space Y if there exist continuous functions $f : Y \to X$ and $g : X \to Y$ such that $f \circ g$ is homotopic to the identity of X.

Definition 2.2.13. Let X be a space dominated by a CW-complex K. We denote by ΔX the minimal dimension among the CW-complexes dominating X (it may be infinite), that is,

$$\Delta X := \min\{\dim K \colon K \text{ is a CW-complex dominating } X\}.$$

The following theorem requires yet more vocabulary to be understood. We present the objects here but urge the reader to turn to any of the references [Whi49a], [MGR21] and [Hat02] for greater depth.

Given a topological space X, its homotopy groups $\pi_n(X)$ are a higher-dimensional generalization of its fundamental group $\pi_1(X)$, formed by the equivalence classes modulo homotopy of loops in X. Every continuous function $f : X \to Y$ induces homomorphisms $f_n : \pi_n(X) \to \pi_n(Y)$ between said homotopy groups, which will be isomorphisms if and only if f is a homotopy equivalence.

Theorem 2.2.14. *Let X and Y be two connected topological spaces dominated by respective CW-complexes. Then a function $f : X \to Y$ is a homotopy equivalence if and only if $f_n : \pi_n(X) \to \pi_n(Y)$ is a group isomorphism for every n such that $1 \le n < N+1$, where $N = max\,\{\Delta X, \Delta Y\}$.*

Let us now proceed with the two lemmas, both with proof. For the proof of the first one we have relied on the proof of Lemma 5 found in [Whi49b, p. 67].

Lemma 2.2.15 (Whitehead's lemma)**.** *Let $\varphi_0, \varphi_1 : \partial D^\lambda \to X$ be two homotopic attaching maps. Then, the identity function of X can be extended to a homotopy equivalence*

$$k : X \cup_{\varphi_0} e^\lambda \to X \cup_{\varphi_1} e^\lambda.$$

Proof. First we define the function k and its inverse l, and afterwards we prove that they do indeed give a homotopy equivalence.

Let $\{\varphi_t : \partial D^\lambda \to X\}$ be a homotopy between φ_0 and φ_1; we define k as follows.

$$\begin{cases} k(x) = x & \text{if} \quad x \in X \\ k(tu) = 2tu & \text{if} \quad 0 \le t \le \frac{1}{2},\ u \in \partial D^\lambda \\ k(tu) = \varphi_{2-2t}(u) & \text{if} \quad \frac{1}{2} \le t \le 1,\ u \in \partial D^\lambda, \end{cases}$$

where tu denotes the usual product of the scalar $t \in [0,1]$ with the unit vector $u \in \partial D^\lambda$, that is, it covers all of D^λ. It is continuous since for $t = \frac{1}{2}$ one has

$$k(u/2) = u = \varphi_1(u) \quad \text{and} \quad u \sim \varphi_1(u) \ \text{ in } \ X \cup_{\varphi_1} e^\lambda.$$

Moreover, it is well defined since it sends representatives of the same equivalence class to the same point. In $X \cup_{\varphi_0} e^\lambda$ the related elements are u and $\varphi_0(u)$ for every u in ∂D^λ, and k satisfies

$$k(u) = \varphi_0(u) \ \text{ and } \ k(\varphi_0(u)) = \varphi_0(u),$$

where the second equality holds since $\varphi_0(u)$ is in X.

Similarly, we construct $l : X \cup_{\varphi_1} e^\lambda \to X \cup_{\varphi_0} e^\lambda$ as

$$\begin{cases} l(x) = x & \text{if} \quad x \in X, \\ l(tu) = 2tu & \text{if} \quad 0 \le t \le \frac{1}{2},\ u \in \partial D^\lambda, \\ l(tu) = \varphi_{2t-1}(u) & \text{if} \quad \frac{1}{2} \le t \le 1,\ u \in \partial D^\lambda. \end{cases}$$

One proves analogously to k that l is continuous and well defined.

Now, let us see what $l \circ k$ looks like and give a homotopy from it to the identity of $X \cup_{\varphi_0} e^\lambda$. The case $k \circ l$ is analogous and is left to the discretion of the reader.

$$\begin{cases} l \circ k(x) = x & \text{if} \quad x \in X, \\ l \circ k(tu) = 4tu & \text{if} \quad 0 \leq t \leq \frac{1}{4},\ u \in \partial D^\lambda, \\ l \circ k(tu) = \varphi_{4t-1}(u) & \text{if} \quad \frac{1}{4} \leq t \leq \frac{1}{2},\ u \in \partial D^\lambda, \\ l \circ k(tu) = \varphi_{2-2t}(u) & \text{if} \quad \frac{1}{2} \leq t \leq 1,\ u \in \partial D^\lambda. \end{cases}$$

The last case follows from the fact that $k(tu) = \varphi_{2-2t}(u)$ if $\frac{1}{2} \leq t \leq 1$ and this belongs to X, hence l leaves it fixed. Let us now see the homotopy; we define

$$\begin{cases} h_s(x) = x & \text{if} \quad x \in X, \quad s \in [0,1], \\ h_s(tu) = (4-3s)tu & \text{if} \quad 0 \leq t \leq \frac{1}{4-3s},\ u \in \partial D^\lambda \\ h_s(tu) = \varphi_{(4-3s)t-1}(u) & \text{if} \quad \frac{1}{4-3s} \leq t \leq \frac{2-s}{4-3s},\ u \in \partial D^\lambda \\ h_s(tu) = \varphi_{\frac{1}{2}(4-3s)(1-t)}(u), & \text{if} \quad \frac{2-s}{4-3s} \leq t \leq 1,\ u \in \partial D^\lambda. \end{cases}$$

It is clear that for $s = 0$ we recover $l \circ k$ and for $s = 1$ one has the identity of $X \cup_{\varphi_0} e^\lambda$. Continuity is easily proven by looking at the behavior at the endpoints of each piece. That h_s is well defined is checked as in the case of k.

□

Lemma 2.2.16. *Let $\varphi : \partial D^\lambda \to X$ be an attaching map. Every homotopy equivalence $f : X \to Y$ can be extended to a homotopy equivalence*

$$F : X \cup_\varphi e^\lambda \to Y \cup_{f \circ \varphi} e^\lambda.$$

Proof. We define F by means of

$$\begin{cases} F|_X = f, \\ F|_{e^\lambda} = \mathrm{id}_{e^\lambda}. \end{cases}$$

Let $g : Y \to X$ be a homotopy inverse of f; we define

$$G : Y \cup_{f \circ \varphi} e^\lambda \to X \cup_{g \circ f \circ \varphi} e^\lambda$$

in the same way as F,

$$\begin{cases} G|_Y = g, \\ F|_{e^\lambda} = \mathrm{id}_{e^\lambda}. \end{cases}$$

Now, since $g \circ f$ is homotopic to the identity of X we have that $g \circ f \circ \varphi$ is homotopic to φ. Hence it follows from the previous lemma that there exists a homotopy equivalence

$$k : X \cup_{g \circ f \circ \varphi} e^\lambda \to X \cup_\varphi e^\lambda.$$

which we construct just as in the proof of said lemma using a homotopy from $g \circ f$ to the identity of X, $\{h_t : X \to X\}$.

Let us first see that the composition $(k \circ G \circ F) : X \cup_\varphi e^\lambda \to X \cup_\varphi e^\lambda$ is homotopic to the identity; from now on we will refer to this composition as kGF.

It is simple to check that kGF is given by

$$\begin{cases} kGF(x) = g(f(x)) & \text{if} \quad x \in X, \\ kGF(tu) = 2tu & \text{if} \quad 0 \leq t \leq \frac{1}{2},\ u \in \partial D^{\lambda}, \\ kGF(tu) = h_{2-2t}(\varphi(u)) & \text{if} \quad \frac{1}{2} \leq t \leq 1,\ u \in \partial D^{\lambda}. \end{cases}$$

Now, we denote by $q_s : X \cup_{\varphi} e^{\lambda} \to X \cup_{\varphi} e^{\lambda}$ the homotopy we need from kGF to the identity. It is given by

$$\begin{cases} q_s(x) = h_s(x) & \text{if} \quad x \in X, \\ q_s(tu) = \frac{2}{1+s}tu & \text{if} \quad 0 \leq t \leq \frac{1+s}{2},\ u \in \partial D^{\lambda}, \\ q_s(tu) = h_{2-2t+s}(\varphi(u)) & \text{if} \quad \frac{1+s}{2} \leq t \leq 1,\ u \in \partial D^{\lambda}. \end{cases}$$

At $s = 0$ we recover kGF since $h_0 \equiv g \circ f$; the other two pieces are clear. At $s = 1$ one has h_1, which is the identity on X, $q_1(tu) = tu$ for all $t \in [0, 1]$, and $q_1(u) = h_1(\varphi(u))$ on the last piece (since t can only take the value 1), which is equal to $\varphi(u)$, which is related to u. One easily checks that there are no continuity problems by observing that the values q takes when $t = \frac{1+s}{2}$ are related. Moreover, q is well defined since $q_s(u) = h_s(\varphi(u)) = q_s(\varphi(u))$.

Thus we have proven that F has a left homotopy inverse. One proves analogously that G has a right homotopy inverse. Now we must prove that kG is, indeed, also a right homotopy inverse of F. For this we prove the following result first:

> If a function $F : X \to Y$ has a left homotopy inverse L and a right homotopy inverse R, then F is a homotopy equivalence and both R and L are two-sided homotopy inverses.

Indeed, the relations $L \circ F \simeq \mathrm{id}_X$ and $F \circ R \simeq \mathrm{id}_Y$ imply

$$L \simeq L(F \circ R) = (L \circ F)(R) \simeq R.$$

Hence, since L is related to R,

$$L \circ F \simeq R \circ F \simeq \mathrm{id}_X.$$

Thus, R is a two-sided homotopy inverse. It is proven for L in the same way.

We finish the proof of the lemma as follows. We know that both F and G have a left homotopy inverse.

Now, since $k(GF) \simeq \mathrm{id}_{X \cup_{\varphi} e^{\lambda}}$ and we know that k has a left homotopy inverse, we have that $(GF)k \simeq \mathrm{id}_{X \cup_{g \circ f \circ \varphi} e^{\lambda}}$. Here we have applied the development of the proof of the intermediate result with $l = L$, $F = k$ and $R = GF$.

In the same way, since $G(Fk) \simeq \mathrm{id}_{X \cup_{g \circ f \circ \varphi} e^{\lambda}}$ and we know that G has a left homotopy inverse, we have $(Fk)G \simeq \mathrm{id}_{Y \cup_{f \circ \varphi} e^{\lambda}}$.

Finally, since $F(kG) \simeq \mathrm{id}_{Y \cup_{f \circ \varphi} e^{\lambda}}$ and F has a left homotopy inverse, we have that F is a homotopy equivalence, which concludes the proof. $\square$

With both lemmas proven, we proceed to state and prove the last result of the section.

Theorem 2.2.17. *If f is a Morse function on a smooth manifold M and M^a is compact for every $a \in \mathbb{R}$. Then M has the homotopy type of a CW-complex with one λ-cell for each critical point of index λ.*

Proof. Let $\{c_i\}_{i\in\mathcal{C}}$ be the set of critical values of f ordered in such a way that $c_1 < c_2 < c_3 < \cdots$ holds. This set cannot have any accumulation point; let us see it.

In the first place, that the set $\{c_i\}_{i\in\mathcal{C}}$ has an accumulation point c implies that there exists a sequence $\{c_{i_k}\}_{k\in\mathbb{N}}$ converging to a value c. Now, because of this, there exist infinitely many c_i smaller than $c+1$ and since, by definition, for each c_i there exists a critical point $p_i \in M$ such that $f(p_i) = c_i$, we have that M^{c+1} contains infinitely many critical points of f.

By the compactness of M^{c+1}, this infinite collection of critical points necessarily contains a sequence $\{p_{i_k}\}_{k\in\mathbb{N}}$ converging to a point p of M^{c+1}. By continuity of f we have that

$$f(p) = \lim_{k\to\infty} f(p_{i_k}) = \lim_{k\to\infty} c_{i_k} = c$$

and by continuity of the differential of f, p is a critical point. This is a contradiction since f is a Morse function, hence p is non-degenerate and, therefore, isolated.

Let $a \in \mathbb{R} \setminus \{c_i\}_{i\in\mathcal{C}}$; we now proceed by induction on the number of critical values smaller than a.

For the base case suppose $a < c_1$. It is clear that M^a is empty since if it were not, being compact, f would have to attain a minimum on M^a and would have a critical value smaller than c_1.

For the induction hypothesis suppose that M^a with $c_{i-1} < a < c_i$ has the homotopy type of a CW-complex K and let us denote by $h : M^a \to K$ a homotopy equivalence. Let us then see that given $a' > c_i$ one has that $M^{a'}$ has the homotopy type of a CW-complex.

By Theorem 2.2.9, for ε sufficiently small $M^{c_i+\varepsilon}$ has the homotopy type of

$$M^{c_i-\varepsilon} \cup_{\varphi_1} e^{\lambda_1} \cup \cdots \cup_{\varphi_m} e^{\lambda_m}$$

for certain attaching maps $\varphi_1 : \partial D^{\lambda_1} \to M^{c_i-\varepsilon}, \ldots, \varphi_m : \partial D^{\lambda_m} \to M^{c_i-\varepsilon}$; let us take $a' = c_i + \varepsilon$. Moreover, by Theorem 2.2.5 there exists a homotopy equivalence $g : M^{c_i-\varepsilon} \to M^a$.

Now, by the cellular approximation theorem (2.2.11) we have that for every $j \in \{1, \ldots, m\}$ $h \circ g \circ \varphi_j$ is homotopic to a function

$$\psi_j : \partial D^{\lambda_j} \to K^{\lambda_j - 1},$$

where K^{λ_j-1} denotes the $(\lambda_j - 1)$-skeleton of K. Thus, by Lemmas 2.2.15 and 2.2.16 we have that

$$K \cup_{\psi_1} e^{\lambda_1} \cup \cdots \cup_{\psi_m} e^{\lambda_m}$$

is a CW-complex and has the same homotopy type as $M^{c_i+\varepsilon}$. We have completed the induction: M^a has the homotopy type of a CW-complex for every a.

If M is compact we are done, since we would have a finite number of critical values and M equals M^a for every $a \geq \max f$, that is, the induction covers all of M.

If M is not compact but the number of critical values is finite, then all the critical points of M lie in one same M^a and, similarly to the proof of Theorem 2.2.5, one can prove that M^a is a deformation retract of M and, once again, we are done.

If we have an infinite number of critical values and, therefore, of critical points, the construction carried out in the induction gives us the following sequence of homotopy equivalences,

$$\begin{array}{ccccccc} M^{a_1} & \subset & M^{a_2} & \subset & M^{a_3} & \subset \cdots \\ \downarrow \varphi_1 & & \downarrow \varphi_2 & & \downarrow \varphi_3 & \\ K_1 & \subset & K_2 & \subset & K_3 & \subset \cdots \end{array}$$

such that each one extends the previous one. Let $K = \bigcup_{i=1}^{\infty} K_i$ with the smallest topology making each of the inclusions $i_j : K_j \hookrightarrow K$ continuous. That is, the topology such that $\mathcal{U} \subseteq K$ is open if and only if for every $j \in \mathbb{N}$ one has that $i_j^{-1}(\mathcal{U})$ is open in K_j.

We define the limit function $\Phi : M \to K$ as the function taking a point $x \in M$ to the point $i_i \circ \varphi_i(x) \in K$. In the first place, it is well defined since the choice of $\varphi_i(x)$ does not matter. Indeed, x belongs to all the M^{a_j} from a certain $j \in \mathbb{N}$ onwards and, since each homotopy equivalence extends the previous one, the image of x under φ_j is the same regardless of the chosen index j.

Moreover, it is continuous since, given $\mathcal{U} \subseteq K$ open, one has

$$\Phi^{-1}(\mathcal{U}) = (i_i \circ \varphi_i)^{-1}(\mathcal{U}) = \varphi_i^{-1} \circ i_i^{-1}(\mathcal{U}) \overset{\text{open}}{\subseteq} M,$$

where the last inclusion follows from the continuity of i_i and φ_i. One has that Φ induces isomorphisms between the homotopy groups in all dimensions, hence, by Theorem 2.2.14, it is a homotopy equivalence, which concludes the proof.

$\square$

Remark 2.2.18. At the end of the proof, in order to see that Theorem 2.2.14 can be applied, one must check that both K and M are dominated by CW-complexes. It is clear that K is dominated by itself. To prove that M is, it suffices to observe that it is a deformation retract of a tubular neighborhood of itself, which is, in turn, diffeomorphic to an open subset of $\mathbb{R}^m$ for some m, and every open subset of $\mathbb{R}^m$ has the homotopy type of a CW-complex.

Remark 2.2.19. We have proven that each M^a has the homotopy type of a CW-complex with a finite number of cells, one for each critical point. This is true even when a is a critical value (see Remark 2.2.8).

2.2.1 Applications of the theorems on the attaching of λ-cells

We briefly present two examples of applications of the theorems about the attaching of λ-cells.

Theorem 2.2.20 (Reeb's theorem)**.** *If M is a compact smooth manifold and f is a Morse function on M with only two critical points, then M is homeomorphic to a sphere.*

Proof. Said critical points must necessarily be the maximum and the minimum. Let us call the critical points p and q and say that $f(p) = 0$ is the minimum and $f(q) = 1$ is the maximum.

We know by the Morse Lemma (1.2.8) that there exist neighborhoods $\mathcal{U}_p$ of p and $\mathcal{U}_q$ of q and local coordinate systems $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ such that p and q have, respectively, coordinates 0 and f is of the form

$$f(x) = x_1^2 + x_2^2 + \cdots + x_n^2 \quad \text{and} \quad f(x) = 1 - y_1^2 - y_2^2 - \cdots - y_n^2$$

Now, let $\varepsilon > 0$ be sufficiently small, we have that

$$M^\varepsilon = \{p \in M \colon f(p) \leq \varepsilon\} \quad \text{and} \quad f^{-1}[1-\varepsilon, 1] = \{p \in M \colon 1 - \varepsilon \leq f(p) \leq 1\}.$$

Both, in local coordinates, correspond to closed n-disks of radius $\sqrt{\varepsilon}$ and center 0; therefore, they are homeomorphic to said balls.

On the other hand, since $f^{-1}[\varepsilon, 1-\varepsilon]$ contains no critical point, by Theorem 2.2.5 M^ε and $M^{1-\varepsilon}$ are diffeomorphic. Thus, M is equal to the union of the two n-disks $M^{1-\varepsilon}$ and $f^{-1}[1-\varepsilon, 1]$ glued along their common boundary, which is homeomorphic to a sphere.

□

Remark 2.2.21.

(i) The result is true even when the critical points are degenerate; however, the proof is more complicated.

(ii) Using Theorem 2.2.17 it is easy to see that M has the homotopy type of a CW-complex with one 0-cell and one n-cell. A sphere admits a CW-complex structure with those cells; however, the theorem does not tell us how they are attached. Hence we cannot ensure that M has the homotopy type of the sphere, which would suffice to prove the result thanks to the *generalized Poincaré conjecture.*

(iii) It is not necessary for M to be diffeomorphic to the sphere. This is due to the existence of *exotic structures* in some dimensions: spaces homeomorphic to the sphere whose differentiable structures are not diffeomorphic to that of the standard sphere.

Another application of the theorems of the previous section is the following. Let $\mathbb{CP}^n$ be the complex projective space of dimension n. We will think of $\mathbb{CP}^n$ as the space whose points are the equivalence classes of $(n+1)$-tuples $(z_0, \dots, z_n)$ of complex numbers, denoted by $[z_0 : z_1 : \cdots : z_n]$, such that $\sum_j |z_j|^2 = 1$.

We define the function $f : \mathbb{CP}^n \to \mathbb{R}$ given by

$$f[z_0 : z_1 : \cdots : z_n] = \sum_{j=0}^{n} c_j |z_j|^2,$$

where $c_0, \dots, c_n \in \mathbb{R}$ are pairwise distinct constants. To find its critical points let us consider the following local coordinate system of $\mathbb{CP}^n$. Let $\mathcal{U}_0$ be the subspace of points such that z_0 is different from 0 and let

$$|z_0| \frac{z_j}{z_0} = x_j + i y_j$$

where $x_1, y_1, \dots, x_n, y_n : \mathcal{U}_0 \to \mathbb{R}$ are coordinate functions taking $\mathcal{U}_0$ diffeomorphically to the unit ball of $\mathbb{R}^{2n}$. It is clear that

$$|z_j|^2 = x_j^2 + y_j^2 \quad \text{and} \quad |z_0|^2 = 1 - \sum_{j=1}^{n} (x_j^2 + y_j^2),$$

where the second equality follows from the fact that $\sum_j |z_j|^2 = 1$ implies $|z_0|^2 = 1 - \sum_{j=1}^{n} |z_j|^2$. And therefore f in these local coordinates is of the form

$$f = c_0 + \sum_{j=1}^{n} (c_j - c_0)(x_j^2 + y_j^2)$$

since

$$\sum_{j=0}^{n} c_j |z_j|^2 = c_0 |z_0|^2 + \sum_{j=1}^{n} c_j |z_j|^2 = c_0 \Big(1 - \sum_{j=1}^{n} (x_j^2 + y_j^2)\Big) + \sum_{j=1}^{n} c_j (x_j^2 + y_j^2).$$

Thus, we obtain that for every $j \in \{1, \dots, n\}$

$$\frac{\partial f}{\partial x_j} = 2x_j (c_j - c_0) \quad \text{and} \quad \frac{\partial f}{\partial y_j} = 2y_j (c_j - c_0),$$

whereby the only point of $\mathcal{U}_0$ at which all the partial derivatives vanish is $p_0 = [1 : 0 : \cdots : 0]$.

Given that $\mathrm{H}_p f$ is a diagonal matrix whose entries are non-zero constants, p_0 is non-degenerate. Moreover, it is simple to see that its index is twice the number of constants c_j that are smaller than c_0.

Analogously we can consider the equivalent coordinate systems for the sets $\mathcal{U}_1, \dots, \mathcal{U}_n$, obtaining the non-degenerate critical points

$$p_1 = [0 : 1 : 0 : \cdots : 0], \dots, p_1 = [0 : 0 : \cdots : 0 : 1]$$

such that the index of p_k is equal to twice the number of constants c_j smaller than c_k. Since the union of the $\mathcal{U}_j$ makes up the whole of $\mathbb{CP}^n$, it follows that these are the only critical points of f, whereby every possible index between 0 and $2n$, that is, the even ones, occurs exactly once.

By Theorem 2.2.17, $\mathbb{CP}^n$ has the homotopy type of a CW-complex of the form

$$e^0 \cup_{\varphi_1} e^2 \cup_{\varphi_2} e^4 \cup_{\varphi_3} \cdots \cup_{\varphi_n} e^{2n}.$$

for certain attaching maps φ_j.

Remark 2.2.22. The decomposition of $\mathbb{CP}^n$ we have just obtained recovers the notion that $\mathbb{CP}^n$ is $\mathbb{C}^n$ with a $\mathbb{CP}^{n-1}$ attached at infinity. Let us see it:

Let $H = \{p = [z_0 : z_1 : \cdots : z_n] \in \mathbb{CP}^n \mid z_0 = 0\}$, then $\mathbb{CP}^n \setminus H$ contains the points whose first coordinate is non-zero, that is, those belonging to the classes with representatives $[1 : z_1 : \cdots : z_n]$. Thus, we have that $\mathbb{CP}^n \setminus H$ is homeomorphic to $\mathbb{C}^n$, which is homeomorphic to a $2n$-cell. Now, H is, in turn, homeomorphic to $\mathbb{CP}^{n-1}$. Whereby $\mathbb{CP}^n$ is the result of attaching a $2n$-cell to H. Iterating this process one arrives at the cell decomposition we had found.

Chapter 3

Introduction to homology

Looking ahead to the next and final chapter on the *Morse inequalities*, we need to introduce the concept of *homology* of a manifold.

In this chapter we develop the theoretical framework necessary to prove the Morse inequalities. In the first place we construct the homology of a space from the simplicial point of view. Afterwards we deal with the particular case of CW-complexes. Finally we define the Betti numbers and the Euler–Poincaré characteristic.

Throughout the development certain technical aspects will be omitted, either for being foreign to our objectives, or for requiring a depth we cannot cover in this format. We invite the reader to turn to the references mentioned if they seek a more rigorous and detailed treatment.

3.1 Simplicial homology

When we want to study topological spaces of low dimension, the fundamental group $\pi_1(X)$ turns out to be very useful. However, it does not allow us to distinguish between spaces of higher dimensions; for example, the fundamental group of $\mathbb{S}^n$ is trivial for $n \geq 2$. This can be solved by considering the higher homotopy groups $\pi_n(X)$. The problem is that these can turn out to be extremely complicated to compute. This idea is reinforced when one realizes that there does not even exist a robust way of computing the homotopy groups of $\mathbb{S}^n$ for arbitrary $n > 1$. We give as an example some of the known homotopy groups of $\mathbb{S}^2$ to get an idea of their complexity:

	π_1	π_2	π_3	π_4	π_5	π_6	π_7	π_8	π_9	π_{10}	π_{11}	π_{12}	π_{13}	π_{14}	π_{15}
$\mathbb{S}^2$	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{12} \times \mathbb{Z}_2$	$\mathbb{Z}_{84} \times \mathbb{Z}_2^2$	$\mathbb{Z}_2^2$

Luckily, the homology groups $H_n(X)$ exist. These are simpler to compute and also resolve limitations of the fundamental group when dealing with spaces of high dimensions. The price to pay for greater computability translates into having to develop a theory that is not only complex but a priori more opaque.

Once again, we are in luck, because for the task at hand the so-called *simplicial homology* suffices. This is a restricted version of the more complicated and general *singular homology* and, although there will be aspects of the latter we must introduce, the vocabulary we need is definitely simpler. The general line we will follow for the development of this theory is the one used in [MGR21], starting from a triangulated topological space.

3.1.1 Triangulations

Triangulating a topological space consists essentially in expressing it as the union of *polyhedra* of its same dimension suitably glued together.

Definition 3.1.1.

(i) A *convex k-polyhedron*, P^k, is the convex hull of a finite number of points of $\mathbb{R}^n$ with non-empty interior.

(ii) An *l-face* of P^k is a subset $P^l \subset \partial P^k$ obtained as the intersection of P^k with a hyperplane that does not meet its interior and which has dimension l

Definition 3.1.2. A *triangulation* of a topological space X is denoted

$$\tau = \{(P^k_\alpha, f_\alpha) : \ 0 \leq k \leq n, \ \alpha \in \Lambda_k\}$$

and consists of:

1. A family of polyhedra P^k_α of dimensions between 0 and n.
2. For each polyhedron, a continuous function $f_\alpha : P^k_\alpha \to X$ with the following properties:
   (a) $f_\alpha : \mathring{P}^k_\alpha \to f_\alpha(\mathring{P}^k_\alpha) \subset X$ is a homeomorphism.
   (b) The sets $C^k_\alpha = f_\alpha(P^k_\alpha) \subset X$ form a locally finite closed cover of X.
   (c) The images $f_\alpha(\mathring{P}^k_\alpha) \subset X$ form a disjoint cover of X.
   (d) For each l-face $P^l \in \partial P^k_\alpha$ there exists a function in the triangulation $f_\beta : P^l_\beta \to X$ such that $f_\beta = f_\alpha \circ L$, where $L : P^l_\beta \to P^l \subset P^k_\alpha$ is an affine isomorphism followed by the inclusion.

Each C^k_α is a k-cell and $n = \dim(X)$ the maximal dimension of the polyhedra of the triangulation.

We recall that a cover $\mathcal{C} = \{C_i\}$ of a topological space X is said to be *locally finite* if every $x \in X$ has an open neighborhood $\mathcal{U}_x$ intersecting only a finite number of elements of $\mathcal{C}$.

We will say that the topology of a space X is *coherent* with respect to a cover $\mathcal{C}$ if the intersection of every open set $\mathcal{U}$ of X with any closed set C_i of the cover is open in C_i.

One has that, given a locally finite closed cover $\mathcal{C}$ of a space X, the topology of X is coherent with respect to $\mathcal{C}$. That is, the topology of a topological space X is coherent with respect to the cover obtained by triangulating. This allows us to reconstruct the topology of X having only a triangulation of it, as the collection of subsets $\mathcal{U} \subseteq X$ such that $\mathcal{U} \cap C^k_\alpha$ is open in C^k_α for any $\alpha \in \Lambda_k$.

Property 2.d determines that the intersections of images of two polyhedra P^k_α and P^t_β of a triangulation are formed by l-cells; they cannot intersect their interiors. Moreover, we allow each function f_α to send more than one l-face to one same l-cell.

Definition 3.1.3. With the notation of the previous definition, a *regular triangulation* is a triangulation such that

1. $f_\alpha : P^k_\alpha \to C^k_\alpha$ is a homeomorphism for every $\alpha \in \Lambda_k$.
2. The P^k_α are *k-simplices*, that is, the convex hull of $(k+1)$ affinely independent points.
3. No two k-cells may share all of their vertices.

Remark 3.1.4. A k-simplex is the simplest possible geometric object of dimension k, the k-dimensional analogue of the triangle. Every k-simplex is a k-polyhedron but not the other way around. In fact, every k-polyhedron can be obtained by suitably gluing k-simplices.

A regular triangulation turns out to be simpler to visualize. In a regular triangulation any two distinct k-simplices either do not touch or share exactly one single l-simplex (with $l < k$) of their boundaries. It represents what one sees when thinking of a mesh of triangles, in the 2-dimensional case.

On the other hand, when carrying the polyhedra of a non-regular triangulation onto a space, identifications of the faces of one same polyhedron may appear. In dimension 2, a triangle can for example turn into a "cone" when two of its edges become glued by the function taking it to the space it triangulates. Resulting in two possible inclusions into said space of the vertex that was not common to both.

Finally, we introduce the notion of orientation of a polyhedron.

Definition 3.1.5. An *orientation* o of a k-polyhedron is an orientation of its interior $\mathring{P}^k$, that is, an orientation of $\mathbb{R}^n$.

Since $\mathring{P}^k$ is an open subset of $\mathbb{R}^k$, its orientation is obtained by fixing one of the two possible orientations of $\mathbb{R}^k$. This orientation o induces an orientation $o|_{P^{n-1}}$ on each of the $(n-1)$-faces $P^{n-1} \subset \partial P^k$.

More concretely, a basis of $\mathbb{R}^k$ belongs to one of two equivalence classes of bases under the following relation: two bases B_1 and B_2 are related if the determinant of the change-of-basis matrix is positive. Orienting $\mathbb{R}^k$ is simply choosing one of these two classes.

On the other hand, given an oriented k-polyhedron P^k, the induced orientation on a $k-1$-polyhedron P^{k-1} of its boundary is again a choice of equivalence class of bases of $\mathbb{R}^{k-1}$, but this time forced by the one inducing it. Suppose the orientation of P^k is the one given by the class $[v_1, v_2, \ldots, v_k]$, then the orientation induced on P^{k-1} will be the one given by the class $[w_2, \ldots, w_k]$ satisfying

$$[\nu, w_2, \ldots, w_k] = [v_1, \ldots, v_k]$$

where ν is a vector tangent to P^k, orthogonal to P^{k-1} and pointing towards the exterior of P^k.

In the 0-dimensional case, that is, the vertices, the orientation will be the choice of sign and we will use the following convention: given an edge a and a vertex v, a induces the orientation $o(v) = \{+\}$ if it ends at v and $o(v) = \{-\}$ if it starts at v. If $a = [v_1, v_2]$ and the order determines the orientation of a, then $o(v_1) = \{-\}$ and $o(v_2) = \{+\}$.

It is worth pointing out that, given a regular triangulation, if two k-polyhedra with the same orientation share a $(k-1)$-polyhedron of their boundaries then they induce opposite orientations on it. It suffices to observe that the corresponding vectors ν_1 and ν_2 are opposite.

There exist other notions of triangulation, such as the *pseudotriangulation*, and a whole theory of the so-called *PL manifolds* which for our purpose are not necessary. For greater depth on this topic, technical details and examples we suggest the reader refer to the original source [MGR21].

We close the triangulations section with a result of Whitehead's collected in [Whi78] that will allow us to speak of the homology of any smooth manifold.

Theorem 3.1.6. *Every n-dimensional smooth manifold M admits a triangulation τ.*

This result corresponds to Theorem 7, located on page 819 of [Whi78]. Whitehead first formulates it for closed manifolds and later, in Section 4 (page 823), generalizes it to open manifolds.

3.1.2 Construction of simplicial homology

A slight generalization of the simplicial complexes that give this theory its name are the Δ-complexes. It is on these that texts such as [Hat02] define singular homology. We, as we have already mentioned, will do it for triangulated topological spaces and, later, for CW-complexes.

Definition 3.1.7. Let the pair (X, τ) be a triangulated topological space, with $\tau = \{(P^k_\alpha, f_\alpha) : \ 0 \leq k \leq n, \ \alpha \in \Lambda_k\}$. We choose arbitrary orientations o^k_α for each polyhedron P^k_α of the triangulation and call $e^k_\alpha = (P^k_\alpha, o^k_\alpha)$ the oriented polyhedron. We define $C^\tau_k(X)$ as the free abelian group on the generators e^k_α

$$\mathbb{Z}\langle e^k_\alpha | \alpha \in \Lambda_k \rangle$$

We also define the following group homomorphisms, which we will call *boundary operators*, $\partial_k : C^\tau_k(X) \to C^\tau_{k-1}(X)$ given by

$$\partial_k(e^k_\alpha) = \sum_{\beta \in \Lambda_k} [e^k_\alpha : e^{k-1}_\beta] e^{k-1}_\beta,$$

where the $[e^k_\alpha : e^{k-1}_\beta]$ will be called *incidence numbers* and are given by $[e^k_\alpha : e^{k-1}_\beta] = \sum \varepsilon_i$ where i runs over all the inclusions $L_i : P^{k-1}_\beta \hookrightarrow \partial P^k_\alpha$ and $\varepsilon_i = \pm 1$ depending on whether or not the orientation o^{k-1}_β coincides with the orientation induced as boundary by $L_i(P^{k-1}_\beta) \subset P^k_\alpha$

Remark 3.1.8.

(i) We define $C_k^\tau(X) = 0$ whenever $k < 0$ or $k > n$ and $\partial_k \equiv 0$ whenever $k \leq 0$ or $k > n$.

(ii) We choose arbitrary orientations for the polyhedra given that choosing one or the other comes down to changing the generator e_α^k into $-e_\alpha^k$.

Without any additional background the reader may be wondering how it is possible to describe explicitly the elements of $C_k^\tau(X)$ and how exactly the boundary operators act. Well, we call *k-chains* the elements of $C_k^\tau(X)$, which are formal sums of polyhedra of τ with integer coefficients where the sign depends on the orientations. That is, we choose an orientation to carry the $+$ sign, and the polyhedra whose orientation is the other possible one will carry the $-$ sign. That is, they are of the form

$$\sum_{i=1}^m z_i e_{\alpha_i}^k \quad \text{with} \quad z_i \in \mathbb{Z} \quad \text{and} \quad \alpha_i \in \Lambda_k,\ \forall i \in \{1, \dots, m\}.$$

Above we have described how the boundary operators ∂_k act on the generators of $C_k^\tau(X)$; however, by linearity, they can be extended to group homomorphisms from $C_k^\tau(X)$ to $C_{k-1}^\tau(X)$. Thus,

$$\partial_k \left(\sum_{i=1}^m z_i e_{\alpha_i}^k \right) = \sum_{i=1}^m z_i \partial_k (e_{\alpha_i}^k).$$

Now, the boundary operator takes a k-chain to the $(k-1)$-chain constituting its boundary taking the orientation into account; in the 2-dimensional case, we would say it takes a 2-polyhedron to the sum of its edges. An indispensable property of the boundary operators is the following.

Proposition 3.1.9. *The boundary operators satisfy $\partial_k \circ \partial_{k+1} = 0$ for all $k \geq 0$.*

Proof. It is possible to prove that every triangulation can be subdivided to obtain a regular triangulation. Bearing in mind that for a regular triangulation the notation and the computations simplify considerably, we are going to prove the result in this case. Since no identifications occur between $(k-1)$-simplices of the boundary of each k-polyhedron when triangulating, we have that there is only one possible inclusion for each one, that is, $[e_\alpha^k : e_\beta^{k-1}] = \pm 1$ always.

Now, it suffices to check that the statement holds for the generators of $C_{k+1}^\tau(X)$. Let e_α^{k+1} be an oriented $(k+1)$-polyhedron generating $C_{k+1}^\tau(X)$, we have that

$$\partial_k \circ \partial_{k+1}(e_\alpha^{k+1}) = \partial_k \left(\sum_{\beta \in \Lambda_{k+1}} [e_\alpha^{k+1} : e_\beta^k] e_\beta^k \right) = \sum_{\gamma \in \Lambda_k} \left(\sum_{\beta \in \Lambda_{k+1}} [e_\alpha^{k+1} : e_\beta^k][e_\beta^k : e_\gamma^{k-1}] \right) e_\gamma^{k-1}$$

Let e_γ^{k-1} be any $(k-1)$-polyhedron included in e_α^{k+1}. Since there is only one possible inclusion of e_γ^{k-1} into ∂e_α^{k+1} , there exist exactly two oriented k-polyhedra e_β^k and $e_{\beta'}^k$ such that

$$e_\gamma^{k-1} \subset e_\beta^k \subset e_\alpha^{k+1} \quad \text{and} \quad e_\gamma^{k-1} \subset e_{\beta'}^k \subset e_\alpha^{k+1}$$

Hence

$$\partial_k \circ \partial_{k+1}(e_\alpha^{k+1}) = \sum_{\gamma \in \Lambda_k} \big([e_\alpha^{k+1} : e_\beta^k][e_\beta^k : e_\gamma^{k-1}] + [e_\alpha^{k+1} : e_{\beta'}^k][e_{\beta'}^k : e_\gamma^{k-1}] \big) e_\gamma^{k-1}$$

Suppose that $[e_\alpha^{k+1} : e_\beta^k] = [e_\alpha^{k+1} : e_{\beta'}^k]$, then $[e_\beta^k : e_\gamma^{k-1}] = -[e_{\beta'}^k : e_\gamma^{k-1}]$ hence the summands cancel.

In the same way, if $[e_\alpha^{k+1} : e_\beta^k] = -[e_\alpha^{k+1} : e_{\beta'}^k]$ then $[e_\beta^k : e_\gamma^{k-1}] = [e_{\beta'}^k : e_\gamma^{k-1}]$ and they cancel again.

□

Remark 3.1.10. We will call *chain complex* any sequence of the form

$$\dots \xrightarrow{\partial_{k+2}} A_{k+1} \xrightarrow{\partial_{k+1}} A_k \xrightarrow{\partial_k} A_{k-1} \xrightarrow{\partial_{k-1}} \dots,$$

where the A_i are abelian groups, as in our case, and $\partial_k \circ \partial_{k+1} = 0$ holds for all $k \geq 0$. Said property is equivalent to $\mathrm{im}(\partial_{k+1}) \subseteq \ker(\partial_k)$. We will say that the sequence is *exact* if equality holds.

We extract from the result that if a k-chain is a boundary, then the boundary of the latter vanishes when taking the orientations into account. Let us imagine a 2-dimensional disk; it is homeomorphic to a 2-simplex, which gives us a triangulation. We have three vertices v_0, v_1 and v_2, three edges $[v_0, v_1]$, $[v_1, v_2]$ and $[v_2, v_0]$ and one face $[v_0, v_1, v_2]$. Hence,

$$\partial_1(\partial_2([v_0, v_1, v_2])) = \partial_1([v_0, v_1] + [v_1, v_2] + [v_2, v_0]) = (v_1 - v_0) + (v_2 - v_1) + (v_0 - v_2) = 0.$$

The boundary of the 2-simplex has no boundary, $[v_0, v_1] + [v_1, v_2] + [v_2, v_0] \in \mathrm{im}(\partial_2) \subseteq \ker(\partial_1)$.

Thus, we can define the k-th homology group of X as follows.

Definition 3.1.11. Let (X, τ) be a triangulated space. The *k-th simplicial homology group* of X is the abelian group $H_k^\tau(X)$ defined by

$$\begin{aligned} Z_k^\tau(X) &= \ker(\partial_k : C_k^\tau(X) \to C_{k-1}^\tau(X)) \\ B_k^\tau(X) &= \mathrm{im}(\partial_{k+1} : C_{k+1}^\tau(X) \to C_k^\tau(X)) \\ H_k^\tau(X) &= Z_k^\tau(X)/B_k^\tau(X). \end{aligned}$$

Remark 3.1.12. Let X be a triangulated space of dimension n, we observe that

- $H_k^\tau(X) = 0$ for all $k > n$, since $C_k^\tau(X) = 0$.
- $H_n^\tau(X)$ is a free abelian group. Indeed, since $C_{n+1}^\tau(X) = 0$ we have that $B_n^\tau(X) = 0$ hence $H_n^\tau(X) = Z_k^\tau(X) \subset C_n^\tau(X)$, which is free for being a subgroup of a free group.
- If X is moreover compact, then $H_k^\tau(X)$ is a finitely generated abelian group. One has that $C_k^\tau(X)$ is isomorphic to $\mathbb{Z}^{n_k}$, where $n_k = |\Lambda_k|$, hence it is finitely generated, and every subgroup of a finitely generated abelian group is in turn finitely generated. Thus, $Z_k^\tau(X)$ and $B_k^\tau(X)$ are finitely generated and the result follows.

Once again, this definition deserves a clarification. The elements of the subgroups $Z_k^\tau(X)$ and $B_k^\tau(X)$ of $C_k^\tau(X)$ will be called, respectively, *cycles* and *boundaries*. The first name is received by the k-chains whose boundary is empty (as in the case of the 2-simplex) and the second by those constituting the boundary of a formal sum of $(k+1)$-chains taking the orientation into account. The elements of $H_k^\tau(X)$ are, therefore, the equivalence classes of cycles whose difference is a boundary.

One would wish that the homology groups did not depend on the chosen triangulation and were the same as those obtained via singular homology. Indeed, this holds.

Theorem 3.1.13. *Let (X, τ) be a triangulated space. Then $H_k^\tau(X)$ is isomorphic to $H_k(X)$ for all $k > 0$. Where $H_k(X)$ denotes the singular homology group of X.*

Once again, we will let the reader find the proof in the original source, by virtue of our decision to avoid formalizing singular homology for our purposes. From now on we will refer to the homology groups as $H_k(X)$, without specifying a triangulation.

Let us finish the simple example of the disk (we will provide more examples for the case of CW-complexes later on). Let D^2 be the disk; then, using the already mentioned triangulation formed by one 2-simplex, one has $C_0(D^2) = \mathbb{Z}\langle v_0, v_1, v_2\rangle \simeq \mathbb{Z}^3$, $C_1(D^2) = \mathbb{Z}\langle [v_0, v_1], [v_1, v_2], [v_2, v_0]\rangle \simeq \mathbb{Z}^3$ and $C_2(D^2) = \mathbb{Z}\langle [v_0, v_1, v_2]\rangle \simeq \mathbb{Z}$.

Now, $Z_2(D^2) = \ker(\partial_2) = \{0\}$, since the elements of $C_2(D^2)$ are of the form $z([v_0, v_1, v_2])$ with z an integer and $\partial_2(z[v_0, v_1, v_2]) = z([v_0, v_1] + [v_1, v_2] + [v_2, v_0]) \neq 0$ if z is non-zero. Moreover, $B_2(D^2) = \{0\}$ necessarily, hence $H_2(D^2) = \{0\}$.

We continue: $B_1(D^2) = \mathrm{im}(\partial_2) = \{z([v_0, v_1]+[v_1, v_2]+[v_2, v_0])\}_{z\in\mathbb{Z}} \simeq \mathbb{Z}$ and $Z_1(D^2) = \ker(\partial_1) = B_1(D^2)$, hence $H_2(D^2) \simeq \mathbb{Z}/\mathbb{Z} = \{0\}$.

There is no need to check that $Z_0(D^2) = \ker(\partial_0) = C_0(D^2) \simeq \mathbb{Z}^3$. Let us see what $B_0(D^2)$ looks like. The images of the generators of $C_1(D^2)$ are

$$\partial_1([v_0, v_1]) = v_1 - v_0, \quad \partial_1([v_1, v_2]) = v_2 - v_1 \quad \text{and} \quad \partial_1([v_2, v_0]) = v_0 - v_2.$$

Hence $B_0(D^2) = \mathbb{Z}\langle v_1 - v_0, v_2 - v_1, v_0 - v_2\rangle$; however, $(v_0 - v_2) = (v_0 - v_1) + (v_1 - v_2)$. Hence in reality one has

$$B_0(D^2) = \mathbb{Z}\langle v_1 - v_0, v_2 - v_0\rangle \simeq \mathbb{Z}^2.$$

Thus, $H_0(D^2) = \mathbb{Z}^3/\mathbb{Z}^2 \simeq \mathbb{Z}$.

What happens if we remove the interior of the 2-simplex? We obtain a circle $\mathbb{S}^1$ triangulated by three 1-simplices and three 0-simplices. Now, the groups C_1 and C_0 remain, as does Z_1; however, $B_1(\mathbb{S}^1)$ is now $\{0\}$ necessarily, whereby $H_1(\mathbb{S}^1) \simeq \mathbb{Z}/\{0\} = \mathbb{Z}$. This coincides with the expected result since now $[v_0, v_1]+[v_1, v_2]+[v_2, v_0]$ is a cycle that is not the boundary of anything, hence it represents a non-trivial class of the homology group.

Before continuing, we list some results that might be of interest to the reader in the context of smooth manifolds and Morse theory. However, we decide not to linger, since it is not central to our objective, and we present them without proof. The proofs can be found in: [MGR21, p. 87] the first one and [MGR21, p. 97] the other three.

1. For any triangulated space X, $H_0(X) = \mathbb{Z}^r$ where r is the number of path-connected components of X.
2. If M is a compact, connected, orientable triangulated n-manifold without boundary, then $H_n(M) = \mathbb{Z}$.
3. If M is a compact, connected, non-orientable triangulated n-manifold without boundary, then $H_n(M) = 0$.
4. If M is a compact, connected n-manifold with (non-empty) boundary, then again $H_n(M) = 0$.

3.2 Cellular homology

We recall that some of the spaces holding special interest for us, together with smooth manifolds, are CW-complexes.

A triangulated space constitutes a CW-complex; after all, each of the C^k_α that are joined to build a triangulation is a k-cell. It is to be expected then that *cellular homology* is constructed in a manner similar to the simplicial one, although now instead of k-polyhedra suitably glued to a space X we will have k-cells attached to one another by the attaching maps.

Let us first introduce some concepts to avoid possible technical gaps in the passage from triangulations to CW-complexes in general.

Definition 3.2.1. Given two chain complexes (A_k, ∂_k) and (B_k, ∂'_k), we call a *chain map* a family f_k of homomorphisms such that $\partial'_k \circ f_k = f_{k-1} \circ \partial_k$ for all k.

This results in the following commutative diagram.

$$\begin{array}{ccccccccc}
\dots & \xrightarrow{\partial_{k+2}} & A_{k+1} & \xrightarrow{\partial_{k+1}} & A_k & \xrightarrow{\partial_k} & A_{k-1} & \xrightarrow{\partial_{k-1}} & \dots \\
 & & f_{k+1} \downarrow & & f_k \downarrow & & f_{k-1} \downarrow & & \\
\dots & \xrightarrow{\partial_{k+2}} & B_{k+1} & \xrightarrow{\partial_{k+1}} & B_k & \xrightarrow{\partial_k} & B_{k-1} & \xrightarrow{\partial_{k-1}} & \dots
\end{array}$$

Definition 3.2.2. Let $f\colon (A_\bullet, \partial) \to (B_\bullet, \partial')$ be a chain map. Since $f \circ \partial = \partial' \circ f$, one has that $f(\ker \partial_k) \subset \ker \partial'_k$ and $f(\operatorname{im} \partial_{k+1}) \subset \operatorname{im} \partial'_{k+1}$. Therefore, f descends to the quotient, giving rise to a homomorphism $f_*\colon H_k(A_\bullet, \partial) \to H_k(B_\bullet, \partial')$ which we call the *induced homomorphism in homology.*

Remark 3.2.3. It is clear that $(g \circ f)_* = g_* \circ f_*$, for morphisms $(A_\bullet, \partial) \xrightarrow{f} (B_\bullet, \partial') \xrightarrow{g} (C_\bullet, \partial'')$.

Definition 3.2.4. Let X be a CW-complex. We orient each of the k-cells of X and define

$$C_k^{CW}(X) = \oplus_{\alpha \in \Lambda_k} \mathbb{Z}\langle D^k_\alpha \rangle,$$

and the boundary operators $\partial_k : C_k^{CW}(X) \to C_{k-1}^{CW}(X)$ are given by

$$\partial_k(D^k_\alpha) = \sum_{\beta \in \Lambda_{k-1}} [D^k_\alpha : D^{k-1}_\beta] D^{k-1}_\beta.$$

Just as in simplicial homology, the elements of $C_k^{CW}(X)$ are formal sums of k-cells with integer coefficients. Now, the boundary operators still take a k-cell to its boundary, only this time said boundary will be (taking the orientation into account) the image of its attaching map.

Let us be clearer: let $i_\alpha(D_\alpha^k)$ be a k-cell of a CW-complex X and $\varphi_\alpha : \partial D_\alpha^k \simeq \mathbb{S}^{k-1} \to X_{k-1}$ its attaching map. Then, $[D_\alpha^k : D_\beta^{k-1}]$ measures how many times (it may be zero) $\varphi_\alpha(\partial D_\alpha^k)$ completely traverses the $(k-1)$-cell $i_\beta(D_\beta^{k-1})$ and in which direction: if the latter coincides with the orientation of $i_\beta(D_\beta^{k-1})$ the sign will be positive and if it differs, negative. Thus, we obtain a formal sum of elements of $(k-1)$-cells, that is, an element of $C_{k-1}^{CW}(X)$.

More formally, the boundary operator ∂_k performs the following journey. The attaching map induces a homomorphism in homology

$$(\varphi_\alpha)_* : H_{k-1}(\partial D_\alpha^k) \to H_{k-1}(X_{k-1}).$$

This homomorphism takes the *fundamental class* $[\partial D_\alpha^k]$, which is essentially the generator of $H_{k-1}(\partial D_\alpha^k) \simeq \mathbb{Z}\langle[\partial D_\alpha^k]\rangle$, to an element of $H_{k-1}(X_{k-1})$. Next, the quotient $q : X_{k-1} \to X_{k-1}/X_{k-2} \simeq \vee_\beta \mathbb{S}_\beta^{k-1}$ induces another homomorphism in homology

$$q_* : H_{k-1}(X_{k-1}) \to H_{k-1}(X_{k-1}/X_{k-2}) = H_{k-1}(\vee_\beta \mathbb{S}_\beta^{k-1}) \simeq \oplus_\beta \mathbb{Z}\langle D_\beta^{k-1}\rangle = C_{k-1}^{CW}(X).$$

In turn, $\partial_k \circ \partial_{k+1} = 0$ also holds. Now that we have seen how the boundary operators act, the homology groups are constructed exactly as in the simplicial case. Given a CW-complex X, the *k-th cellular homology group* of X is the abelian group $H_k^{CW}(X)$ defined by

$$\begin{aligned} Z_k^{CW}(X) &= \ker(\partial_k : C_k^{CW}(X) \to C_{k-1}^{CW}(X)) \\ B_k^{CW}(X) &= \operatorname{im}(\partial_{k+1} : C_{k+1}^{CW}(X) \to C_k^{CW}(X)) \\ H_k^{CW}(X) &= Z_k^{CW}(X)/B_k^{CW}(X). \end{aligned}$$

Once more we refer the reader to the original source [MGR21] if they wish to clarify some of the finer technical details we are not going to prove here. It can be proven that $H_k(X) \simeq H_k^{CW}(X)$ for any triangulated space X. Thus, by transitivity, we have that the homology groups of a space are isomorphic regardless of the technique we apply to compute them. For this reason, from now on we will refer to the homology groups solely as $H_k(X)$.

It also holds that the homology groups are invariant under homeomorphisms and, in fact, under homotopy equivalences ([Hat02, p. 111]). We state only the second result by virtue of the fact that the homotopy type is also invariant under homeomorphisms.

Theorem 3.2.5. *The homomorphisms $f_* : H_k(X) \to H_k(Y)$ induced in homology by a homotopy equivalence $f : X \to Y$ between two topological spaces are isomorphisms.*

The result follows from the fact that two homotopic functions induce the same homomorphisms in homology; therefore, a homotopy equivalence and its inverse induce inverse homomorphisms, hence isomorphisms.

We finish with a few examples of how to compute the homology of a CW-complex.

Example 3.2.6.

(i) We can endow $\mathbb{S}^n$ with a CW-complex structure with one 0-cell and one n-cell by identifying the boundary of an n-disk to the point.

Thus, it is simple to see that $H_0(\mathbb{S}^n) \simeq \mathbb{Z}$ since $\mathbb{S}^n$ is path-connected, and that $H_1(\mathbb{S}^n), \dots, H_{n-1}(\mathbb{S}^n)$ are all $\{0\}$ as the chain groups in those dimensions are empty.

Now, $H_n(\mathbb{S}^n) \simeq \mathbb{Z}$ necessarily since $Z_n(\mathbb{S}^n) = C_n(\mathbb{S}^n) \simeq \mathbb{Z}$ as $\partial_n \equiv 0$, and $B_n(\mathbb{S}^n) = \{0\}$ by construction of the chain complex.

(ii) The torus T^2 can be considered a CW-complex with one 0-cell e^0, two 1-cells e_1^1 and e_2^1 and one 2-cell e^2 in the following way. Take the usual construction of the flat torus, a square whose parallel sides have been identified. The boundary of said square is the wedge sum of two circles, $\mathbb{S}^1 \vee \mathbb{S}^1$, that is, two tangent circles; these will be our 1-cells, the point of tangency our 0-cell. The 2-cell will be the interior of the square.

Now, $C_2(T^2) \simeq \mathbb{Z}\langle e^2 \rangle \simeq \mathbb{Z}$, $C_1(T^2) \simeq \mathbb{Z}\langle e_1^1 \rangle \oplus \mathbb{Z}\langle e_2^1 \rangle \simeq \mathbb{Z}^2$ and $C_0(T^2) \simeq \mathbb{Z}\langle e^0 \rangle \simeq \mathbb{Z}$.

The 2-cell e^2 is attached to the 1-cells along the loop $e_1^1 e_2^1 (e_1^1)^{-1} (e_2^1)^{-1}$ where the exponent -1 means it is traversed in the opposite direction. In terms of chains this reads

$$\partial_2(e^2) = e_1^1 + e_2^1 - e_1^1 - e_2^1 = 0.$$

Hence we are left with $Z_2(T^2) = C_2(T^2) \simeq \mathbb{Z}$ and $B_1(T^2) = \{0\}$.

Both 1-cells are attached to the 0-skeleton at e^0; both are loops, hence

$$\partial_1(e_1^1) = \partial_1(e_2^1) = e^0 - e^0 = 0.$$

Whereby $Z_1(T^2) = C_1(T^2) \simeq \mathbb{Z}^2$. In this way, one has

$$H_2(T^2) = Z_2/B_2 = \mathbb{Z}/\{0\} = \mathbb{Z} \quad \text{and} \quad H_1(T^2) = Z_1/B_1 = \mathbb{Z}^2/\{0\} = \mathbb{Z}^2.$$

And $H_0(T^2) \simeq \mathbb{Z}$ since it is path-connected.

(iii) By the CW-complex structure given for $\mathbb{CP}^n$ in Subsection 2.2.1 one has the following chain complex

$$0 \to C_{2n} \simeq \mathbb{Z} \to C_{2n-1} \simeq \{0\} \to \cdots \to C_2 \simeq \mathbb{Z} \to C_1 \simeq \{0\} \to C_0 \simeq \mathbb{Z} \to 0$$

where for the even dimensions the chain groups are isomorphic to $\mathbb{Z}$ and for the odd ones to $\{0\}$. This means that $Z_k = C_k$ and $B_k = \{0\}$ for all k, hence the homology groups of $\mathbb{CP}^n$ come out as

$$H_k(\mathbb{CP}^n) = \begin{cases} \mathbb{Z} & \text{if } k \leq 2n \text{ is even,} \\ \{0\} & \text{if } k \text{ takes other values.} \end{cases}$$

3.3 Relative homology

We conclude the development of the aspects of homology we need with *relative homology*. In essence, in relative homology what we do is ignore all the chains of a subspace. The construction is similar to the ones we already know.

Definition 3.3.1. Let X be a topological space and $A \subset X$ a subspace. We define $C_k(X, A)$ as the quotient group

$$C_k(X)/C_k(A),$$

hence the chains of A become trivial.

For the boundary operators let us go step by step. In the first place, for defining them to make sense it must happen that ∂_k takes $C_k(A)$ to $C_{k-1}(A)$. In the case of a triangulated space, A must be a *triangulated subspace*, that is, a subspace covered only by complete polyhedra of the original triangulation. For CW-complexes the subspace must be a *subcomplex*, that is, a collection of complete cells of the space.

Once we have this, the boundary operator in relative homology is nothing more than the boundary operator in homology followed by the passage to the quotient. Indeed, thanks to the fact that $\partial_k(C_k(A))$ is contained in $C_{k-1}(A)$, these induce a boundary operator on the quotient given by

$$\partial_k^{rel} = q_{k-1} \circ \partial_k,$$

where the q_k form the family of quotient maps. We will omit the “rel” label whenever the context allows it.

We observe that in this case $\partial_{k-1}^{rel} \circ \partial_k^{rel} = 0$ also holds, since they vanish before passing to the quotient, hence the q_k form a chain map.

Once we have the chain groups and the boundary operators defined, the *relative homology groups* are constructed in the same way as in cellular and simplicial homology. That is, given a triangulated topological space or a CW-complex X and a triangulated subspace or subcomplex A, the *k-th relative homology group* of the pair (X, A) is the abelian group $H_k(X, A)$ constructed as follows

$$\begin{aligned}
Z_k(X, A) &= \ker(\partial_k^{rel} : C_k(X, A) \to C_{k-1}(X, A)) \\
B_k(X, A) &= \operatorname{im}(\partial_{k+1}^{rel} : C_{k+1}(X, A) \to C_k(X, A)) \\
H_k(X, A) &= Z_k(X, A)/B_k(X, A).
\end{aligned}$$

Note. As one would expect, if the subspace A is for example contractible or empty, the non-relative homology groups are recovered.

Proposition 3.3.2. *For any pair (X, A), with X either a triangulated topological space or a CW-complex and A a triangulated subspace or a subcomplex respectively, one has the following exact sequence*

$$\cdots \longrightarrow H_n(A) \longrightarrow H_n(X) \longrightarrow H_n(X, A) \longrightarrow H_{n-1}(A) \longrightarrow H_{n-1}(X) \longrightarrow \cdots \longrightarrow H_0(X, A) \longrightarrow 0.$$

Proving this is one of the main objectives of the relative homology section of [Hat02, p. 115], on which we have based this section.

An example of the usefulness of this exact sequence for computing the relative homology groups is the following.

Example 3.3.3. Let the n-disk D^n and its boundary $\mathbb{S}^{n-1}$ be given; let us compute the relative homology groups $H_k(D^n, \mathbb{S}^{n-1})$ for $n \geq 2$. By the previous proposition, we have the following exact sequence,

$$\begin{tikzcd}
\cdots \arrow[r] & H_k(\mathbb{S}^{n-1}) \arrow[r] & H_k(D^n) \simeq \{0\} \arrow[r] & H_k(D^n, \mathbb{S}^{n-1}) \arrow[dlll, out=0, in=180] \\
H_{k-1}(\mathbb{S}^{n-1}) \arrow[r] & H_{k-1}(D^n) \simeq \{0\} \arrow[r] & H_{k-1}(D^n, \mathbb{S}^{n-1}) \arrow[r] & \cdots
\end{tikzcd}$$

We extract from it the following short exact sequence,

$$\{0\} \xrightarrow{f} H_k(D^n, \mathbb{S}^{n-1}) \xrightarrow{g} H_{k-1}(\mathbb{S}^{n-1}) \xrightarrow{h} \{0\},$$

and one has

$$\{0\} = \operatorname{im} f = \ker g \quad \text{and} \quad \operatorname{im} g = \ker h = H_{k-1}(\mathbb{S}^{n-1}).$$

Hence g is injective and we arrive at

$$H_k(D^n, \mathbb{S}^{n-1}) \simeq \operatorname{im} g = H_{k-1}(\mathbb{S}^{n-1}) \simeq \begin{cases} \mathbb{Z} & \text{if} \quad k = n, \\ 0 & \text{if} \quad k \neq n, \ k \geq 2. \end{cases}$$

For the cases $k = 2$ and $k = 1$ we look at the end of the sequence.

$$\begin{tikzcd}
\cdots \arrow[r] & H_1(\mathbb{S}^{n-1}) \arrow[r] & H_1(D^n) \simeq \{0\} \arrow[r] & H_1(D^n, \mathbb{S}^{n-1}) \arrow[dlll, bend left] \\
H_0(\mathbb{S}^{n-1}) \simeq \mathbb{Z} \arrow[r] & H_0(D^n) \simeq \mathbb{Z} \arrow[r] & H_0(D^n, \mathbb{S}^{n-1}) \arrow[r] & \{0\}
\end{tikzcd}$$

One has that $H_0(\mathbb{S}^{n-1}) \simeq \mathbb{Z} \to H_0(D^n) \simeq \mathbb{Z}$ is the isomorphism induced by the inclusion. We extract again the part between $\{0\}$ and $\{0\}$,

$$\{0\} \xrightarrow{f} H_1(D^n, \mathbb{S}^{n-1}) \xrightarrow{g} \mathbb{Z} \xrightarrow{h} \mathbb{Z} \xrightarrow{i} H_0(D^n, \mathbb{S}^{n-1}) \xrightarrow{j} \{0\}.$$

We observe that $\{0\} = \operatorname{im} f = \ker g$ and, h being an isomorphism, also $\operatorname{im} g = \ker h = \{0\}$. Hence g is injective and its image is $\{0\}$, so $H_1(D^n, \mathbb{S}^{n-1}) \simeq \{0\}$. Now,

$$\ker i = \operatorname{im} h = \mathbb{Z} \quad \text{and} \quad \operatorname{im} i = \ker j = H_0(D^n, \mathbb{S}^{n-1}).$$

Hence $\{0\} = \mathbb{Z}/\mathbb{Z} \simeq H_0(D^n, \mathbb{S}^{n-1})$.

For the case $n = 1$ it happens that $\mathbb{S}^0$ has two connected components, hence $H_0(\mathbb{S}^0) = \mathbb{Z}^2$. To see how $H_1(D^1, \mathbb{S}^0)$ and $H_0(D^1, \mathbb{S}^0)$ turn out, it suffices to observe that $C_1(D^1, \mathbb{S}^0) \simeq \mathbb{Z}$ and $C_0(D^1, \mathbb{S}^0) \simeq \{0\}$. Therefore,

$$Z_1(D^1, \mathbb{S}^0) \simeq \mathbb{Z}, \quad Z_0(D^1, \mathbb{S}^0) = \{0\} \quad \text{and} \quad B_1(D^1, \mathbb{S}^0) = \{0\}.$$

Thus, we obtain $H_1(D^1, \mathbb{S}^0) = \mathbb{Z}$ and $H_0(D^1, \mathbb{S}^0) = \{0\}$, which fits with what was obtained for the higher-dimensional cases.

We present one last result that we will need to prove the Morse inequalities: the excision theorem [Hat02, p. 119]. It describes when the relative homology groups $H_k(X, A)$ are unaffected by removing a subspace $Z \subset A$.

Theorem 3.3.4 (Excision theorem)**.** *Let X be a topological space and $A, B \subset X$ subspaces whose interiors cover X. Then, the inclusion $(B, A \cap B) \hookrightarrow (X, A)$ induces isomorphisms $H_n(B, A \cap B) \to H_n(X, A)$ for all n.*

Just as explained in [Hat02], the proof of the excision theorem requires a certain technical detour involving a construction known as barycentric subdivision. It is worth knowing the result but we will not cover the proof here.

3.4 Other invariants related to homology

We briefly introduce two topological invariants related to the homology of a space that will be central in the last section.

Definition 3.4.1. Let X be a topological space. We define the *k-th Betti number of X* as

$$b_k(X) = \operatorname{rank} H_k(X).$$

For the general definition of the rank of an abelian group we refer the reader to page 85 of [Fuc70]. In the case where the homology groups are finitely generated, as is that of a compact space, their rank is determined by their decomposition according to the *structure theorem for finitely generated abelian groups*. Thus, rank $H_k(X)$ is r in the decomposition

$$H_k(X) = \mathbb{Z}^r \oplus \mathbb{Z}_{d_1} \oplus \cdots \oplus \mathbb{Z}_{d_m},$$

where $d_1, \ldots, d_m$ are integers such that $d_1|d_2|\ldots|d_m$.

Definition 3.4.2. Let X be a topological space whose homology groups are finitely generated and such that $H_k(X) = 0$ for all $k > N$. The *Euler–Poincaré characteristic of X* is

$$\chi(X) = \sum_{k=0}^{N} (-1)^k b_k(X).$$

Note. A usual definition of the Euler–Poincaré characteristic in terms of the polyhedra of a triangulation can be found on page 39 of [MGR21]. In it, X is required to be compact so that summing over the polyhedra of the triangulation makes sense; here we impose a weaker condition.

Given that in our case we do not always have a triangulation at hand, it is convenient to take the more general version above in terms of the Betti numbers.

Both for the Betti numbers and for the Euler–Poincaré characteristic one can change $b_k(X)$ and $\chi(X)$ into their relative versions $b_k(X, A)$ and $\chi(X, A)$ and the definitions carry over to the relative homology groups.

Chapter 4

Morse inequalities

When Morse developed his theory, throughout the twenties [Bot80], the mathematical scene was very different; it was not until 1949 that Whitehead even formalized the concept of CW-complex as we know it [Kle00]. In particular, Theorem 2.2.17 had not yet been proven, so the relation between the topology of a smooth manifold and the critical points of a Morse function was described by means of a collection of inequalities.

In this chapter we present this original formulation following once again the development found in [Mil63].

4.1 Subadditivity

We begin by introducing some results on subadditive functions.

Throughout this chapter we will refer to pairs of spaces (X, Y). It is worth clarifying that these are not arbitrary pairs of spaces, but pairs such that Y is a subspace of X for which it makes sense to construct the relative homology groups (see Section 3.3). Let us also recall that in the case where Y is empty or contractible we recover the homology of X.

Definition 4.1.1. Let S be a function that takes pairs (X, Y) of topological spaces and assigns to them an integer $S(X, Y) \in \mathbb{Z}$. We say that S is *subadditive* if whenever $X \supset Y \supset Z$ one has

$$S(X, Z) \leq S(X, Y) + S(Y, Z).$$

If under the same hypotheses equality always holds we will say that S is *additive*.

Example 4.1.2.

(i) The function $b_k(X, Y)$ assigning to each pair (X, Y) its k-th Betti number is subadditive (as long as the Betti numbers involved are finite). Indeed, let X, Y and Z be such that $X \supset Y \supset Z$; one has the following exact sequence:

$$\cdots \longrightarrow H_k(Y, Z) \xrightarrow{f} H_k(X, Z) \xrightarrow{g} H_k(X, Y) \xrightarrow{h} \cdots$$

This arises from generalizing the sequence mentioned in 3.3.2 to a triple [Hat02, p. 118]. From it the subadditivity is deduced; let us see how. Given a homomorphism between abelian groups of finite rank $\varphi : G \to H$, by the first isomorphism theorem, $G/\ker\varphi$ is isomorphic to $\mathrm{im}\varphi$, hence one has

$$\text{rank } G = \text{rank}(\ker\ \varphi) + \text{rank}(\text{im } \varphi).$$

In our case, we have the following equalities

$$\text{im } f = \ker\ g \quad \text{and} \quad \text{im } g = \ker\ h.$$

Moreover,

$$\begin{aligned} b_k(Y,Z) &= \text{rank}(\ker\ f) + \text{rank}(\text{im } f) = \text{rank}(\ker\ f) + \text{rank}(\ker\ g), \\ b_k(X,Z) &= \text{rank}(\ker\ g) + \text{rank}(\text{im } g), \\ b_k(X,Y) &= \text{rank}(\ker\ h) + \text{rank}(\text{im } h) = \text{rank}(\text{im } g) + \text{rank}(\text{im } h). \end{aligned}$$

Whereby

$$b_k(X,Z) - b_k(Y,Z) = \text{rank}(\text{im } g) - \text{rank}(\ker\ f) \leq b_k(X,Y).$$

From which the subadditivity is deduced.

(ii) The function assigning to each pair its Euler–Poincaré characteristic $\chi(X,Y)$ is additive. We will not include the proof in this example; we refer the reader to Remark 4.1.5.

Lemma 4.1.3. *Let S be a subadditive function and let $X_0 \subset \cdots \subset X_n$. Then,*

$$S(X_n, X_0) \leq \sum_{i=1}^{n} S(X_i, X_{i-1}).$$

If S is additive equality holds.

Proof. We proceed by induction on n. For $n = 1$ equality holds trivially and for $n = 2$ one has the definition of subadditivity. Suppose it holds for $n-1$, then

$$S(X_{n-1}, X_0) \leq \sum_{i=1}^{n-1} S(X_i, X_{i-1}).$$

Using the subadditivity of S and the induction hypothesis one has

$$S(X_n, X_0) \leq S(X_{n-1}, X_0) + S(X_n, X_{n-1}) \leq \sum_{i=1}^{n} S(X_i, X_{i-1}).$$

□

Lemma 4.1.4. *The function S_k is subadditive, where*

$$S_k(X,Y) = \sum_{i=0}^{k} (-1)^i b_{k-i}(X,Y).$$

Proof. Suppose we have the following exact sequence

$$\cdots \xrightarrow{f_{k+1}} A_k \xrightarrow{f_k} A_{k-1} \xrightarrow{f_{k-1}} A_{k-2} \xrightarrow{f_{k-2}} \cdots \xrightarrow{f_1} A_0 \xrightarrow{f_0} 0.$$

We recall that the rank of each f_i is defined as the rank of im f_i. Thus,

$$\begin{aligned}
&\text{rank } f_{k+1} = \text{rank } (\text{im } f_{k+1}),\\
&\text{rank } A_k = \text{rank } (\text{ker } f_k) + \text{rank } (\text{im } f_k) = \text{rank } (\text{im } f_{k+1}) + \text{rank } f_k.
\end{aligned}$$

From which it follows that

$$\text{rank } f_{k+1} = \text{rank } A_k - \text{rank } f_k.$$

If we keep expanding the ranks of the functions in the same way we arrive at the following

$$\begin{aligned}
\text{rank } f_{k+1} &= \text{rank } A_k - \text{rank } f_k =\\
&= \text{rank } A_k - \text{rank } A_{k-1} + \text{rank } f_{k-1} =\\
&= \text{rank } A_k - \text{rank } A_{k-1} + \text{rank } A_{k-2} - \text{rank } f_{k-2} =\\
&\vdots\\
&= \text{rank } A_k - \text{rank } A_{k-1} + \text{rank } A_{k-2} - \cdots \pm \text{rank } A_0.
\end{aligned}$$

It is clear that the last expression is greater than zero. We now consider the exact sequence of the triple $X \supset Y \supset Z$ (see 4.1.2(i)),

$$\cdots \longrightarrow H_{k+1}(Y,Z) \longrightarrow H_{k+1}(X,Z) \longrightarrow H_{k+1}(X,Y) \xrightarrow{f} H_k(Y,Z) \longrightarrow \cdots$$

and we apply to f what we have just proven. We observe that

$$\text{rank} f = b_k(Y,Z) - b_k(X,Z) + b_k(X,Y) - b_{k-1}(Y,Z) + \cdots \geq 0.$$

Grouping the terms suitably we arrive at

$$S_k(Y,Z) - S_k(X,Z) + S_k(X,Y) \geq 0, \tag{4.1}$$

which concludes the proof.

□

Remark 4.1.5. Taking k sufficiently large one recovers from S_k the Euler–Poincaré characteristic, that is, $S_k(X,Y) = \chi(X,Y)$. Moreover, for this k one has that 4.1 is an equality, which proves the additivity of the Euler–Poincaré characteristic.

4.2 The inequalities

Theorem 4.2.1 (Morse inequalities, weak version)**.** *Let f be a Morse function on a compact n-dimensional manifold M and C_λ the number of critical points of index λ of f. Then,*

$$b_\lambda(M) \leq C_\lambda, \quad \textit{for all } \lambda \in \Lambda, \quad \textit{and} \tag{1_λ}$$

$$\chi(M) = \sum_{\lambda=0}^{n} (-1)^\lambda C_\lambda. \tag{2}$$

Proof. Since M is compact, f has a finite number of critical points. Let $a_1 < \cdots < a_k$ be such that for every $i \in \{1, \ldots, k\}$ one has that M^{a_i} contains exactly i critical points, and $M^{a_k} = M$. Then,

$$H_j(M^{a_i}, M^{a_{i-1}}) = H_j(M^{a_{i-1}} \cup e^{\lambda_i}, M^{a_{i-1}}) = H_j(e^{\lambda_i}, \partial e^{\lambda_i}) = \begin{cases} \mathbb{Z} & \text{if} \quad j = \lambda_i, \\ 0 & \text{in any other case.} \end{cases}$$

The first equality follows from Theorem 2.2.7, where λ_i is the index of the critical point. The second is obtained by applying the excision theorem (3.3.4) to the space $M^{a_{i-1}} \cup e^{\lambda_i}$ and the subspaces $M^{a_{i-1}}$ and e^{λ_i}. The third equality was proven in Example 3.3.3.

Now, applying Lemma 4.1.3 with $M^{a_0} = \emptyset \subset \cdots \subset M^{a_k} = M$ and $S = b_\lambda$ one has

$$b_\lambda(M) \leq \sum_{i=1}^{k} b_\lambda(M^{a_i}, M^{a_{i-1}}) = C_\lambda.$$

If instead we apply the lemma with $S = \chi$ one arrives at

$$\chi(M) = \sum_{i=1}^{k} \chi(M^{a_i}, M^{a_{i-1}}) = \sum_{i=1}^{k} \left(\sum_{\lambda=0}^{n} (-1)^\lambda b_\lambda(M^{a_i}, M^{a_{i-1}}) \right) = \sum_{\lambda=0}^{n} (-1)^\lambda \sum_{i=1}^{k} b_\lambda(M^{a_i}, M^{a_{i-1}}) =$$
$$= \sum_{\lambda=0}^{n} (-1)^\lambda C_\lambda$$

□

We have proven a first relation: the rank of the λ-th homology group of M must be less than or equal to the number of critical points of index λ of any Morse function defined on M. Moreover, given a Morse function, it suffices to find the number of critical points of each possible index to know its Euler–Poincaré characteristic.

Now, one can obtain inequalities slightly stronger than these:

Theorem 4.2.2 (Morse inequalities)**.** *Let f be a Morse function on a compact n-dimensional manifold M and S_λ the function defined in Lemma 4.1.4; for every $\lambda \in \mathbb{Z}^+$ the following inequalities hold*

$$S_\lambda(M) \leq \sum_{i=1}^{k} S_\lambda(M^{a_i}, M^{a_{i-1}}) = C_\lambda - C_{\lambda-1} + \cdots \pm C_0,$$

where $(M^{a_i}, M^{a_{i-1}})$ are as in the proof of 4.2.1. Equivalently,

$$b_\lambda(M) - b_{\lambda-1} + \cdots \pm b_0(M) \leq C_\lambda - C_{\lambda-1} + \cdots \pm C_0. \tag{3_λ}$$

Proof. It suffices to apply the subadditivity of S_λ and Lemma 4.1.3 with the spaces $M^{a_0} = \emptyset \subset \cdots \subset M^{a_k} = M$.

□

It is clear that this collection of inequalities is stronger than the previous one. Indeed, it suffices to observe that adding (3_λ) and ($3_{\lambda-1}$) one obtains inequality (1_λ). On the other hand, if λ is greater than n one has that (3_λ) is the equality (2).

Let us see a way of using these inequalities. Suppose that $C_{\lambda+1} = 0$; then, necessarily $b_{\lambda+1} = 0$. Thus, (3_λ) and ($3_{\lambda+1}$) become

$$\begin{aligned} b_\lambda - b_{\lambda-1} + \cdots \pm b_0 &\leq C_\lambda - C_{\lambda-1} + \cdots \pm C_0, \\ -b_\lambda + b_{\lambda-1} - \cdots \pm b_0(M) &\leq -C_\lambda + C_{\lambda-1} - \cdots \pm C_0. \end{aligned}$$

From which one deduces

$$b_\lambda(M) - b_{\lambda-1} + \cdots \pm b_0(M) = C_\lambda - C_{\lambda-1} + \cdots \pm C_0.$$

Suppose now that $C_{\lambda-1} = 0$; one then has that $b_{\lambda-1} = 0$, and in the same way one arrives at the equality

$$b_{\lambda-2}(M) - b_{\lambda-3} + \cdots \pm b_0(M) = C_{\lambda-2} - C_{\lambda-3} + \cdots \pm C_0.$$

Subtracting now the latter from the first equality one obtains that $b_\lambda = C_\lambda$. We have just proven the following corollary.

Corollary 4.2.3. *If $C_{\lambda+1} = C_{\lambda-1} = 0$ then $b_\lambda = C_\lambda$ and $b_{\lambda+1} = b_{\lambda-1} = 0$.*

We conclude the chapter by applying this last result to the computation of the homology groups of $\mathbb{S}^n$ with $n \geq 2$ and of $\mathbb{CP}^n$, already computed in Example 3.2.6.

Let us recall that $\mathbb{S}^n$ admits a Morse function with exactly two critical points of indices 0 and n (2.2.20). Thus,

$$C_\lambda = \begin{cases} 1, & \text{if } \lambda \in \{0, n\}, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, applying the previous corollary one has that $b_\lambda(\mathbb{S}^n) = 1$ if $\lambda \in \{0, n\}$ and 0 otherwise. Hence we recover the homology groups of $\mathbb{S}^n$,

$$H_k(\mathbb{S}^n) = \begin{cases} \mathbb{Z}, & \text{if } k \in \{0, n\}, \\ \{0\}, & \text{otherwise.} \end{cases}$$

On the other hand, by what was seen at the end of Section 2.2.1, $\mathbb{CP}^n$ admits a Morse function with $n+1$ critical points whose indices are the even numbers from 0 to $2n$. That is,

$$C_\lambda = \begin{cases} 1, & \text{if } \lambda \text{ is even and } \lambda \leq 2n, \\ 0, & \text{otherwise.} \end{cases}$$

Hence for every even index we find ourselves within the hypotheses of the corollary. If λ is even, one has that $C_{\lambda+1} = C_{\lambda-1} = 0$. Hence,

$$b_\lambda(\mathbb{CP}^n) = \begin{cases} 1, & \text{if } \lambda \text{ is even and } \lambda \leq 2n, \\ 0, & \text{otherwise.} \end{cases}$$

Whereby, once again, we recover its homology groups,

$$H_k(\mathbb{CP}^n) = \begin{cases} \mathbb{Z} & \text{if } k \text{ is even and } k \leq 2n, \\ \{0\} & \text{otherwise.} \end{cases}$$

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